

Spectral measures with the original operator domain retained

A sharp spectral cutoff can be constructed without an eigenbasis. This companion builds the scalar measures, the bounded positive-operator calculus, and the exact passage back to a lower-bounded self-adjoint operator. Every product keeps its actual operator domain. The construction applies to the unchanged weighted Hilbert space in the elliptic course, including noncompact domains.

The first two parts supply the measure and Hilbert foundations used by the operator proof. The third constructs the full calculus from positive Bernstein polynomial weights. The fourth recovers the original lower-bounded operator, its recursive powers, and both endpoints of a sharp cutoff. The fifth gives continuous and atomic models with complete domain calculations. The sixth proves the Hilbert geometry, square roots, closed-operator polar decomposition tensor completion, complete direct sums and geometric inverses needed by the trace, operator and divergence arguments. Inner products are linear in the first variable throughout.

1. Constructing the scalar measure from a positive functional

The space throughout is $K = [0, 1]$, with its usual metric and relative topology. All Borel sets and all supports below are relative to K . A positive complex-linear functional means a complex-linear map $L : C(K; \mathbb{C}) \rightarrow \mathbb{C}$ such that $L(f)$ is a nonnegative real number whenever f is real and nonnegative. Inner products and all sesquilinear forms are linear in their first argument and conjugate-linear in their second argument.

This proof constructs the measure, proves countable additivity and regularity, develops the integration facts used, and then proves the exact complex-measure statements for operator pairings. Its measure-theoretic input is set theory and elementary compactness of the interval; it does not use a representation theorem or a spectral theorem. Completeness of a Hilbert space is used only in the final operator-representation step, whose proof is also included.

1.1 Positivity and the mass constant

Let L be positive, and put $M = L(1) \geq 0$. For real continuous u , the functions $u_+ = \max(u, 0)$ and $u_- = \max(-u, 0)$ are continuous and nonnegative, so $L(u) = L(u_+) - L(u_-)$ is real. Consequently $L(\bar{f}) = \overline{L(f)}$ for complex continuous f . If $u \leq v$ are real, positivity gives $L(u) \leq L(v)$.

For complex f , if $L(f) \neq 0$, choose a scalar ζ of modulus one with $\zeta L(f) = |L(f)|$. Then $\operatorname{Re}(\zeta f) \leq |f|$ pointwise, and complex linearity and the preceding real-valued property give

$$|L(f)| = L(\operatorname{Re}(\zeta f)) \leq L(|f|) \leq M\|f\|_\infty. \quad (\text{PR1})$$

The same inequality holds when $L(f) = 0$. Thus positivity implies boundedness, $\|L\| \leq M$, and evaluation on 1 gives $\|L\| = M$. In particular $M = 0$ implies $L = 0$. This case is included in every construction below.

1.2 Mass assigned to open sets

For each open $U \subset K$, define

$$c(U) = \sup\{L(f) : f \in C(K; \mathbb{R}), 0 \leq f \leq 1, \operatorname{supp} f \subset U\}. \quad (\text{PR2})$$

The zero function is allowed, and the empty support causes no exception. We have $0 \leq c(U) \leq M$, $c(\emptyset) = 0$, $c(K) = M$, and c is monotone under inclusion.

We first prove the finite decomposition needed for countable subadditivity. Let $S \subset \bigcup_{i=1}^r U_i$ be a nonempty compact set, with each U_i open. Put $d_i(x) = \text{dist}(x, K \setminus U_i)$ when $U_i \neq K$, and put $d_i = 1$ when $U_i = K$. These functions are continuous, nonnegative, and positive at exactly the points of U_i . The continuous function $\max_i d_i$ is positive on S , so its minimum there is positive. Choose $\delta > 0$ smaller than that minimum and also smaller than one. Set

$$h_i = (d_i - \delta/2)_+, \quad h_0(x) = \text{dist}(x, S), \quad \psi_i = \frac{h_i}{h_0 + \sum_{j=1}^r h_j}. \quad (\text{PR3})$$

The denominator is positive on S by the choice of δ , and positive off S because $h_0 > 0$ there. Thus each ψ_i is continuous and between zero and one. Its support is contained in the compact set $\{d_i \geq \delta/2\} \subset U_i$. On S , the sum of the ψ_i is one.

Suppose that $U \subset \bigcup_{j=1}^\infty U_j$, with all these sets open, and take an admissible f in (PR2) for U . If $f = 0$, there is nothing to prove. Otherwise its compact support S has a finite subcover from the U_j . Apply (PR3) to that subcover and set $f_i = f\psi_i$. Each f_i is admissible for its corresponding U_i , and $\sum_i f_i = f$: on S the partition sums to one, and off S the function f is zero. Therefore

$$L(f) = \sum_i L(f_i) \leq \sum_{j=1}^\infty c(U_j). \quad \text{Hence} \quad c(U) \leq \sum_{j=1}^\infty c(U_j). \quad (\text{PR4})$$

For finitely many pairwise disjoint open sets $U_1, \dots, U_r$, admissible functions f_i have disjoint supports, so $0 \leq \sum_i f_i \leq 1$ and its support is contained in $\bigcup_i U_i$. Approximating each supremum in (PR2) within an arbitrarily small positive error gives $c(\bigcup_i U_i) \geq \sum_i c(U_i)$. Combined with (PR4), this proves equality. For countably many disjoint open sets, monotonicity gives the lower bound by every finite partial sum, while (PR4) gives the upper bound. Thus c is countably additive on disjoint open unions.

1.3 Outer measure and separation by positive distance

For every subset $E \subset K$, define

$$\mu^*(E) = \inf\{c(U) : U \subset K \text{ open, } E \subset U\}. \quad (\text{PR5})$$

The family is never empty, since K is allowed. The function μ^* is monotone, vanishes on the empty set, and is at most M . For subsets E_j and $\epsilon > 0$, choose open $U_j \supset E_j$ with $c(U_j) \leq \mu^*(E_j) + \epsilon 2^{-j}$, where indices begin at one. By (PR4),

$$\mu^*\left(\bigcup_j E_j\right) \leq c\left(\bigcup_j U_j\right) \leq \sum_j \mu^*(E_j) + \epsilon. \quad (\text{PR6})$$

Letting $\epsilon \downarrow 0$ proves countable subadditivity. Thus μ^* is an outer measure. For open U , (PR5) and monotonicity of c show $\mu^*(U) = c(U)$; in particular $\mu^*(K) = M$.

If nonempty subsets A, B have distance $d > 0$, take any open $U \supset A \cup B$ and form $U_A = U \cap \{x : \text{dist}(x, A) < d/3\}$ and $U_B = U \cap \{x : \text{dist}(x, B) < d/3\}$. They are disjoint open supersets of A, B . To see disjointness, a point in both would admit points of A, B whose distance is less than $2d/3$ plus arbitrarily small errors, contradicting the definition of d . Open-set additivity and monotonicity give

$c(U) \geq c(U_A) + c(U_B) \geq \mu^*(A) + \mu^*(B)$. Take the infimum over U and use outer subadditivity for the reverse inequality. The result, also immediate if one set is empty, is

$$\text{dist}(A, B) > 0 \implies \mu^*(A \cup B) = \mu^*(A) + \mu^*(B). \quad (\text{PR7})$$

1.4 Borel measurability and countable additivity

Call a set $A \subset K$ measurable for μ^* when, for every subset $E \subset K$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A). \quad (\text{PR8})$$

First every closed set F satisfies (PR8). The cases $F = \emptyset, K$ are immediate. Otherwise the function $d_F(x) = \text{dist}(x, F)$ is continuous and satisfies $|d_F(x) - d_F(y)| \leq |x - y|$, by the triangle inequality and taking infima. Fix any subset E , and put

$$E_i = \{x \in E : d_F(x) \geq 1/i\}, \quad A_j = \{x \in E : 1/(j+1) \leq d_F(x) < 1/j\}. \quad (\text{PR9})$$

The set E_i has distance at least $1/i$ from $E \cap F$. Hence (PR7) and monotonicity give $\mu^*(E) \geq \mu^*(E \cap F) + \mu^*(E_i)$. The sets A_j with indices of one fixed parity are separated from one another by positive distance in each finite collection: for indices $j, j+2$, the gap between the indicated distance intervals is at least $1/(j+1) - 1/(j+2) > 0$, and the distance function is 1-Lipschitz. Iterating (PR7) over a finite collection of even indices, and then over odd indices, shows that each of the two series of their outer masses has sum at most M . Therefore $\sum_{j \geq 1} \mu^*(A_j)$ converges and its tails tend to zero.

The set $E \setminus F$ is contained in $E_i \cup \bigcup_{j \geq i} A_j$. Outer subadditivity now gives

$$\mu^*(E \setminus F) \leq \mu^*(E_i) + \sum_{j \geq i} \mu^*(A_j). \quad (\text{PR10})$$

Combining the preceding two inequalities and letting $i \rightarrow \infty$ proves $\mu^*(E) \geq \mu^*(E \cap F) + \mu^*(E \setminus F)$. Outer subadditivity gives the opposite inequality. This proves (PR8) for every closed F , without assuming continuity from below for an outer measure.

Here is the complete algebra argument for the measurable sets. Complementation preserves (PR8). If A, B satisfy (PR8), split E first by A , then split $E \setminus A$ by B . This gives $\mu^*(E) = \mu^*(E \cap A) + \mu^*((E \setminus A) \cap B) + \mu^*(E \setminus (A \cup B))$. The first two terms are at least $\mu^*(E \cap (A \cup B))$ by subadditivity. Together with the reverse subadditivity bound, this proves (PR8) for $A \cup B$. Thus the measurable sets form an algebra.

For pairwise disjoint measurable A_j , repeated splitting gives, with $A = \bigcup_j A_j$,

$$\mu^*(E) \geq \sum_{j=1}^r \mu^*(E \cap A_j) + \mu^*(E \setminus A) \quad (r \geq 1). \quad (\text{PR11})$$

Let $r \rightarrow \infty$. The sum is at least $\mu^*(E \cap A)$ by countable subadditivity. Again the reverse inequality is automatic, so A is measurable. Any countable union can be made disjoint by replacing its terms by their differences from the finite preceding union; those differences belong to the algebra. Consequently the measurable sets form a sigma-algebra. Since they contain every closed set, they contain every Borel set.

For disjoint measurable A_j , take $E = \bigcup_j A_j$ in (PR11) and compare with countable subadditivity. The result is exact countable additivity. Restriction to Borel sets therefore gives a finite positive Borel measure

$$\mu(E) = \mu^*(E) \quad (E \text{ Borel}), \quad \mu(K) = M, \quad \mu(U) = c(U) \quad (U \text{ open}). \quad (\text{PR12})$$

No representation or integration theorem has been used to obtain this measure.

1.5 Outer and inner regularity

Equation (PR5), together with (PR12), gives for every Borel set E

$$\mu(E) = \inf_{U \supset E, U \text{ open}} \mu(U). \quad (\text{PR13})$$

To prove inner regularity, choose open $V \supset K \setminus E$ with $\mu(V) < \mu(K \setminus E) + \epsilon$, and put $F = K \setminus V$. Then F is compact and contained in E . Finite additivity on the disjoint parts of V gives $\mu(E \setminus F) = \mu(E \cap V) = \mu(V) - \mu(K \setminus E) < \epsilon$. It follows that

$$\mu(E) = \sup_{F \subset E, F \text{ compact}} \mu(F). \quad (\text{PR14})$$

Both inequalities also hold for empty sets and for zero mass. This is the claimed regularity on every Borel set, rather than only on open sets.

1.6 Integration and the convergence statements actually used

The following construction and proofs apply to any finite positive measure ρ on a sigma-algebra, including μ on the Borel sets. A nonnegative simple function can be written as $s = \sum_{i=1}^r a_i 1_{E_i}$ with disjoint measurable E_i and $a_i \geq 0$. Define $\int s d\rho = \sum_i a_i \rho(E_i)$. Refining two finite partitions by their intersections proves independence of the representation, additivity for simple functions, and monotonicity. For a nonnegative measurable function f , define

$$\int f d\rho = \sup \left\{ \int s d\rho : 0 \leq s \leq f, s \text{ simple} \right\}. \quad (\text{PR15})$$

Countable additivity first implies continuity from below for sets. If $E_k \uparrow E$, the disjoint sets $E_1, E_2 \setminus E_1, \dots$ partition E , so $\rho(E_k) \uparrow \rho(E)$. Taking complements inside a set of finite mass also proves continuity from above.

If $0 \leq f_k \uparrow f$, monotonicity gives $\lim_k \int f_k \leq \int f$. For the reverse inequality, take a nonnegative simple $s \leq f$ and $0 < c < 1$. On every point where $s > 0$, the increasing sequence f_k eventually exceeds cs . Thus the sets $E_k = \{f_k \geq cs\}$ increase to the entire underlying space (points where $s = 0$ already belong to every E_k). Continuity from below for each set in the finite simple partition gives $\int s 1_{E_k} \rightarrow \int s$, while $f_k \geq cs 1_{E_k}$. Hence $\lim_k \int f_k \geq c \int s$. Take the supremum over s and then let $c \uparrow 1$. This proves monotone convergence, including infinite integrals:

$$0 \leq f_k \uparrow f \implies \int f_k d\rho \uparrow \int f d\rho. \quad (\text{PR16})$$

Every nonnegative measurable f has the increasing simple approximants $s_k = 2^{-k} \lfloor 2^k \min(f, k) \rfloor$. They are simple because the cap is finite, are measurable because their level sets are measurable, are increasing because the caps increase and each grid refines its predecessor, and tend pointwise to f . Applying (PR16) to these approximants for f , g , and their sums proves $\int(f+g) = \int f + \int g$. Positive homogeneity follows in the same way or directly from (PR15).

For real f with $\int |f| < \infty$, define $\int f = \int f_+ - \int f_-$. To prove additivity for real integrable f, g , use the pointwise identity $f_+ + g_+ + (f+g)_- = f_- + g_- + (f+g)_+$. All terms have finite integrals because $|f+g| \leq |f| + |g|$; additivity for nonnegative functions and rearrangement therefore give $\int(f+g) = \int f + \int g$. Positive homogeneity is already proved, and $(-f)_+ = f_-$, $(-f)_- = f_+$ proves the negative-scalar case. For complex integrable f , define its integral by its real and imaginary parts; real linearity and multiplication by i prove complex linearity. If its integral is nonzero, multiply by a constant phase making that integral positive real and use $\operatorname{Re}(\zeta f) \leq |f|$, as in (PR1). This proves

$$\left| \int f d\rho \right| \leq \int |f| d\rho. \quad (\text{PR17})$$

Fatou's inequality follows directly from (PR16): for nonnegative measurable f_k , set $h_j = \inf_{k \geq j} f_k$. These measurable functions increase to $\liminf f_k$, and $\int h_j \leq \inf_{k \geq j} \int f_k$. Therefore

$$\int \liminf_k f_k d\rho \leq \liminf_k \int f_k d\rho. \quad (\text{PR18})$$

If complex measurable $f_k \rightarrow f$ pointwise and $|f_k| \leq g$ with $\int g < \infty$, then $|f| \leq g$. Apply (PR18) to the nonnegative functions $2g - |f_k - f|$, whose limit is $2g$. Their integrals are $2 \int g - \int |f_k - f|$, so the resulting inequality forces $\limsup_k \int |f_k - f| \leq 0$. Thus

$$\int |f_k - f| d\rho \rightarrow 0, \quad \int f_k d\rho \rightarrow \int f d\rho. \quad (\text{PR19})$$

The same proof applies to convergence outside a measurable null set by changing the functions to zero there. In particular, on a finite measure space, pointwise convergence of a uniformly bounded sequence of bounded Borel functions implies convergence of their integrals. All monotone and dominated convergence uses below refer to the proofs (PR16)–(PR19), not to a spectral-theoretic convergence assertion.

1.7 Recovery of the original functional

Let $f \in C(K; \mathbb{R})$ be nonnegative, and let $B = \|f\|_\infty$. If $B = 0$, the assertion is immediate. Otherwise fix $\delta > 0$, let $r = \lceil B/\delta \rceil$, and put $U_i = \{f > i\delta\}$, $1 \leq i \leq r$. For each i , choose any function g_i admissible in (PR2) for U_i . At a point x , the number of indices with $g_i(x) > 0$ is at most the number of positive integers i for which $i\delta < f(x)$. Therefore $0 \leq \delta \sum_i g_i(x) \leq f(x)$. Positivity and approximation of the finitely many suprema give

$$L(f) \geq \delta \sum_{i=1}^r c(U_i) = \int s_\delta d\mu, \quad s_\delta = \delta \sum_{i=1}^r 1_{\{f > i\delta\}}. \quad (\text{PR20})$$

The approximation step is finite: choose each $L(g_i)$ within $\epsilon/(r\delta)$ of $c(U_i)$, and then let $\epsilon \downarrow 0$. Pointwise $0 \leq f - s_\delta \leq \delta$, including values of f that are exact multiples of δ . By the integration facts already proved, $\int s_\delta \geq \int f - \delta M$. Let $\delta \downarrow 0$; then $L(f) \geq \int f$.

Apply this same inequality to the nonnegative continuous function $B - f$. Since $\mu(K) = M = L(1)$, it gives $BM - L(f) \geq BM - \int f$, hence $L(f) \leq \int f$. Equality follows. Real continuous functions are differences of their continuous positive and negative parts, and complex continuous functions have real and imaginary parts. Linearity therefore proves

$$L(f) = \int_K f d\mu \quad (f \in C(K; \mathbb{C})), \quad \mu(K) = L(1). \quad (\text{PR21})$$

This proves existence of the representing finite positive regular Borel measure for every positive bounded complex-linear functional.

1.8 Uniqueness and lower semicontinuous approximation

For an open proper nonempty $U \subset K$, the continuous functions $g_k(x) = \min(1, (k \operatorname{dist}(x, K \setminus U) - 1)_+)$ increase to 1_U . They are between zero and one, and each support lies in $\{x : \operatorname{dist}(x, K \setminus U) \geq 1/k\} \subset U$. For $U = K$ use the constant sequence one, and for $U = \emptyset$ use zero. If another finite positive Borel measure ν has the integrals (PR21), monotone convergence for both measures gives $\nu(U) = \lim_k \int g_k d\nu = \lim_k L(g_k) = \mu(U)$.

We give the elementary set argument extending this equality to all Borel sets. A family $\mathcal{D}$ containing K , closed under complements in K , and closed under disjoint countable unions is called a lambda-system. It is also closed under differences $A \setminus B$ when $B \subset A$ and both sets belong to it: take the complement of the disjoint union $(K \setminus A) \cup B$. A family $\mathcal{P}$ closed under finite intersections is a pi-system. If a lambda-system contains a pi-system $\mathcal{P}$ that contains K , it contains the sigma-algebra generated by $\mathcal{P}$. Here is a proof. Let $\mathcal{D}_0$ be the smallest lambda-system containing $\mathcal{P}$. For fixed $A \in \mathcal{P}$, the sets B satisfying $A \cap B \in \mathcal{D}_0$ form a lambda-system: the complement step uses the nested-difference property inside A . This lambda-system contains $\mathcal{P}$, so it contains $\mathcal{D}_0$. Next fix $B \in \mathcal{D}_0$ and repeat the argument with the sets A satisfying $A \cap B \in \mathcal{D}_0$. The first step shows this new lambda-system contains $\mathcal{P}$, hence $\mathcal{D}_0$. Thus $\mathcal{D}_0$ is closed under finite intersections. Complements and finite intersections give finite unions and differences, and disjointizing any countable union then proves it is a sigma-algebra. This proves the assertion.

The Borel sets E with $\mu(E) = \nu(E)$ form a lambda-system: total masses agree, complementation subtracts from that common finite mass, and disjoint countable unions use countable additivity. The open sets are a pi-system containing K and generating the Borel sigma-algebra. The preceding argument proves $\mu = \nu$ on every Borel set. No regularity assumption on the competing measure was needed.

For later use, let $f : K \rightarrow [0, B]$ be lower semicontinuous. For every positive integer k , set

$$f_k(x) = \inf_{y \in K} (f(y) + k|x - y|). \quad (\text{PR22})$$

The infimum is attained: a minimizing sequence has a convergent subsequence in compact K , and lower semicontinuity gives a minimizing limit. The inequality $||x - y| - |x' - y|| \leq |x - x'|$ implies $|f_k(x) - f_k(x')| \leq k|x - x'|$ by comparing infima in both directions. Thus f_k is continuous. It satisfies $0 \leq f_k \leq f$ and increases with k . If y_k is a minimizer for a fixed x , then $k|x - y_k| \leq f_k(x) \leq f(x) \leq B$, so $y_k \rightarrow x$. Lower semicontinuity yields $\liminf_k f_k(x) \geq \liminf_k f(y_k) \geq f(x)$. Together with $f_k \leq f$, this proves $f_k \uparrow f$. Lower semicontinuity makes f Borel because its strict upper level sets are open. Consequently, by (PR16),

$$\int f d\mu = \sup\{L(g) : g \in C(K; \mathbb{R}), 0 \leq g \leq f\}. \quad (\text{PR23})$$

The inequality from right to left is monotonicity; the reverse inequality follows by choosing the functions (PR22).

1.9 Finite complex measures, variation, and integration

The complex measures needed here will be finite linear combinations $\nu = \sum_{r=1}^s c_r \rho_r$ of finite positive Borel measures. They are countably additive because every summand is. Define the positive measure $\lambda = \sum_r |c_r| \rho_r$; then $|\nu(E)| \leq \lambda(E)$. Define total variation by finite measurable partitions:

$$|\nu|(E) = \sup\left\{\sum_{j=1}^r |\nu(E_j)| : E = \bigsqcup_{j=1}^r E_j, E_j \text{ Borel}\right\}. \quad (\text{PR24})$$

It satisfies $|\nu|(E) \leq \lambda(E)$. It is a measure, as follows directly. For disjoint Borel sets E_i and $E = \bigcup_i E_i$, combine near-maximizing finite partitions of the first finitely many E_i , together with the remaining subset of E , to obtain $|\nu|(E) \geq \sum_{i=1}^r |\nu|(E_i)$ for every r . In the other direction, for any finite partition $E = \bigsqcup_j F_j$, countable additivity and triangle inequality give $\sum_j |\nu(F_j)| \leq \sum_i \sum_j |\nu(F_j \cap E_i)| \leq \sum_i |\nu|(E_i)$. Take the supremum over the partition and then let $r \rightarrow \infty$ in the first inequality. This proves countable additivity of $|\nu|$.

For complex simple $s = \sum_j z_j 1_{E_j}$ on a finite disjoint partition, define $\int s d\nu = \sum_j z_j \nu(E_j)$. Refinement proves independence of the partition, and (PR24) gives $|\int s d\nu| \leq \int |s| d|\nu|$. Every bounded complex Borel function has uniformly convergent simple approximants, obtained by rounding its real and imaginary parts to successively finer finite grids on their bounded ranges. The preceding inequality makes their integrals Cauchy, with a limit independent of the approximating sequence. This defines $\int f d\nu$, proves linearity, and gives

$$\left|\int f d\nu\right| \leq \int |f| d|\nu|, \quad \int f d\nu = \sum_{r=1}^s c_r \int f d\rho_r \quad (f \text{ bounded Borel}). \quad (\text{PR25})$$

Both assertions follow first for simple functions, then by uniform approximation; $||s| - |f|| \leq |s - f|$ handles the absolute-value integral. If $f_k \rightarrow f$ pointwise and all functions are uniformly bounded, (PR19) for $|\nu|$ and (PR25) show convergence of the complex integrals. More generally, if $|f_k| \leq g$ with g integrable for $|\nu|$, define integrals of such functions by truncation; the bound (PR25) makes the truncations Cauchy because the absolute-value tail integrals tend to zero by (PR19). The same bound and (PR19) prove dominated convergence in that case as well.

If the positive measures ρ_r are regular, so is λ : for a Borel E , choose compact subsets $F_r \subset E$ and open supersets $U_r \supset E$ making each $|c_r| \rho_r(U_r \setminus F_r)$ as small as prescribed, ignoring zero coefficients. The compact union $F = \bigcup_r F_r$ is contained in E , and the open intersection $U = \bigcap_r U_r$ contains E . Since $U \setminus F \subset U_r \setminus F_r$ for every r , their weighted sum can be made arbitrarily small. As $|\nu| \leq \lambda$, the same F, U prove regularity of $|\nu|$, and $|\nu(E) - \nu(F)| \leq |\nu|(E \setminus F)$, $|\nu(U) - \nu(E)| \leq |\nu|(U \setminus E)$ give the corresponding approximation for ν .

The uniqueness argument extends to any finite complex Borel measure of finite total variation. Indeed, once finiteness of (PR24) is given, the proof that variation is a measure and the integration construction above apply unchanged; the finite positive-measure decomposition was used to establish finiteness and

regularity, not in those subsequent proofs. If $\int f d\nu = 0$ for every continuous f , the continuous open-set approximants in Section 1.8 and dominated convergence for $|\nu|$ give $\nu(U) = 0$ for every open U . The family of Borel sets on which ν vanishes is a lambda-system, since $\nu(K) = 0$, and countable additivity handles disjoint unions. The pi-system argument already proved therefore gives $\nu = 0$. Applying this to a difference proves uniqueness whenever two finite complex measures have equal continuous integrals.

2. Complex measure pairings and Hilbert representation

2.1 Polarization with the linear-first convention

Let H be a complex Hilbert space. Suppose $F_f(x, y)$, for continuous complex f , is complex-linear in f , linear in x , conjugate-linear in y , and satisfies

$$F_f(x, x) \geq 0 \quad \text{when } f \geq 0, \quad 0 \leq F_1(x, x) \leq C\|x\|^2 \quad (x \in H), \quad (\text{PR26})$$

for a fixed finite $C \geq 0$. These are precisely the measure-stage hypotheses, for example when $F_f(x, y) = \langle \Phi(f)x, y \rangle$ and Φ is a positive linear map with $\|\Phi(1)\| \leq C$. A unital continuous $*$ -homomorphism has $C = 1$: for real $f \geq 0$, its continuous square root gives $\Phi(f) = \Phi(\sqrt{f})^* \Phi(\sqrt{f}) \geq 0$, and $\Phi(1) = I$.

For each x , Section 1.1 and Section 1.7 give a unique finite positive regular measure μ_x such that

$$F_f(x, x) = \int f d\mu_x, \quad \mu_x(K) = F_1(x, x) \leq C\|x\|^2. \quad (\text{PR27})$$

For any sesquilinear form B with the linear-first convention, writing $q(z) = B(z, z)$ gives the algebraic identity

$$B(x, y) = \frac{1}{4} \sum_{k=0}^3 i^k q(x + i^k y) = \frac{1}{4} (q(x + y) - q(x - y) + iq(x + iy) - iq(x - iy)). \quad (\text{PR28})$$

This identity does not require B to be Hermitian. Indeed, for $t = i^k$, $q(x + ty) = q(x) + q(y) + \bar{t}B(x, y) + tB(y, x)$. After multiplication by t and summation, the constant terms vanish because $\sum t = 0$, the $B(y, x)$ term vanishes because $\sum t^2 = 0$, and the $B(x, y)$ term is multiplied by $\sum |t|^2 = 4$. This proves the formula and in particular fixes the plus sign before $iq(x + iy)$.

Define the finite regular complex measure

$$\mu_{x,y} = \frac{1}{4} (\mu_{x+y} - \mu_{x-y} + i\mu_{x+iy} - i\mu_{x-iy}). \quad (\text{PR29})$$

Equation (PR25), followed by (PR28) applied to F_f , proves

$$\int f d\mu_{x,y} = F_f(x, y) \quad (f \in C(K; \mathbb{C})). \quad (\text{PR30})$$

The complex-measure uniqueness proved in Section 1.9 now shows that $(x, y) \mapsto \mu_{x,y}$ is sesquilinear as a measure-valued map. For example, the continuous integrals of $\mu_{ax+bz,y} - a\mu_{x,y} - b\mu_{z,y}$ vanish for all f , so this measure is zero; the second variable is treated with conjugated scalars. The same uniqueness gives $\mu_{x,x} = \mu_x$. This step proves sesquilinearity of the actual measures; it is not inferred just from the appearance of the polarization expression.

Fix a Borel set E . Then $B_E(x, y) = \mu_{x,y}(E)$ is a sesquilinear form with nonnegative real diagonal $B_E(x, x) = \mu_x(E)$. Every such form is Hermitian and satisfies Cauchy–Schwarz. For completeness, reality of $B_E(x + ty, x + ty)$ for real t gives that $B_E(x, y) + B_E(y, x)$ is real; reality for it gives that $i(B_E(y, x) - B_E(x, y))$ is real. Together these two equalities imply $B_E(y, x) = \overline{B_E(x, y)}$. If $B_E(y, y) > 0$, substitute $t = -B_E(x, y)/B_E(y, y)$ into the nonnegative quadratic expression for $B_E(x + ty, x + ty)$. Its value is $B_E(x, x) - |B_E(x, y)|^2/B_E(y, y)$, proving the inequality. If $B_E(y, y) = 0$ and $B_E(x, y) \neq 0$, substituting $t = -sB_E(x, y)$ with large positive real s makes that quadratic expression negative, a contradiction. Thus

$$\mu_{y,x}(E) = \overline{\mu_{x,y}(E)}, \quad |\mu_{x,y}(E)|^2 \leq \mu_x(E)\mu_y(E). \quad (\text{PR31})$$

For a finite partition $E = \bigsqcup_j E_j$, use (PR31), then the finite scalar Cauchy–Schwarz inequality, to obtain $\sum_j |\mu_{x,y}(E_j)| \leq \sum_j \sqrt{\mu_x(E_j)\mu_y(E_j)} \leq \sqrt{\mu_x(E)\mu_y(E)}$. The scalar inequality itself follows from nonnegativity of $\sum_j |u_j + tv_j|^2$ by the same quadratic minimization just used. Taking the supremum over partitions yields the sharper local variation estimate

$$|\mu_{x,y}|(E) \leq \sqrt{\mu_x(E)\mu_y(E)}, \quad |\mu_{x,y}|(K) \leq C\|x\|\|y\|. \quad (\text{PR32})$$

In particular this estimate handles zero vectors and zero diagonal mass exactly, without division by their norms or masses.

2.2 Bounded Borel pairings and their convergence

For bounded complex Borel g , define

$$B_g(x, y) = \int g d\mu_{x,y}. \quad (\text{PR33})$$

Measure-valued sesquilinearity and integration linearity show that this is a sesquilinear form. By (PR25) and (PR32),

$$|B_g(x, y)| \leq C\|g\|_\infty\|x\|\|y\|. \quad (\text{PR34})$$

It agrees with F_g for continuous g , and $B_g(x, x) = \int g d\mu_x$. It is positive when $g \geq 0$, and (PR31) gives $B_{\overline{g}}(x, y) = \overline{B_g(y, x)}$.

One additional estimate is useful for strong convergence. For a simple function $g = \sum_j z_j 1_{E_j}$ on a finite disjoint partition of K , (PR31) and finite Cauchy–Schwarz give $|B_g(x, y)| \leq \sum_j |z_j| \sqrt{\mu_x(E_j)\mu_y(E_j)} \leq (\sum_j |z_j|^2 \mu_x(E_j))^{1/2} \mu_y(K)^{1/2}$. Uniform simple approximation extends this to every bounded Borel g : uniform convergence of g_j implies uniform convergence of $|g_j|^2$ because the sequence is uniformly bounded. Therefore

$$|B_g(x, y)|^2 \leq \left(\int |g|^2 d\mu_x \right) \mu_y(K) \leq C\|y\|^2 \int |g|^2 d\mu_x. \quad (\text{PR35})$$

If g_k are uniformly bounded Borel functions and converge pointwise to g , dominated convergence for $|\mu_{x,y}|$ proves $B_{g_k}(x, y) \rightarrow B_g(x, y)$ for every x, y . In fact (PR35) applied to $g_k - g$, with (PR19) for μ_x , proves

$$\sup_{\|y\| \leq 1} |B_{g_k - g}(x, y)| \rightarrow 0 \quad (x \in H). \quad (\text{PR36})$$

For $0 \leq g_k \uparrow g$ with a common finite bound, (PR16) also gives the increasing convergence $B_{g_k}(x, x) \uparrow B_g(x, x)$. For bounded nonnegative lower semicontinuous g , the explicit continuous functions (PR22) give $B_g(x, x) = \sup\{F_f(x, x) : f \in C(K; \mathbb{R}), 0 \leq f \leq g\}$ by (PR23). These conclusions specify the convergence and lower semicontinuous supremum used in the operator application.

2.3 The corresponding bounded operators, with the Hilbert-space step proved

We first prove the Hilbert representation needed to turn (PR33) into an operator. Let ℓ be a bounded conjugate-linear functional on a complex Hilbert space with the linear-first convention. If it is zero, its representing vector is zero. Otherwise $F(y) = \overline{\ell(y)}$ is a nonzero bounded linear functional. The closed affine set $A = \{y : F(y) = 1\}$ is nonempty and has distance $d > 0$ from the origin, since $1 \leq \|F\| \|y\|$ on it. Choose $v_k \in A$ with $\|v_k\| \rightarrow d$. Its midpoints remain in A , and the parallelogram identity yields

$$\|v_k - v_l\|^2 = 2\|v_k\|^2 + 2\|v_l\|^2 - 4\|(v_k + v_l)/2\|^2 \leq 2\|v_k\|^2 + 2\|v_l\|^2 - 4d^2 \rightarrow 0. \quad (\text{PR37})$$

Completeness gives a limit $v \in A$ of norm d . For every $w \in \ker F$, the vector $v + tw$ lies in A for every complex t . Expanding its squared norm and using the minimality of $\|v\|$ for real t and for purely imaginary t proves $\langle w, v \rangle = 0$. Since $y - F(y)v \in \ker F$, it follows that $\langle y, v \rangle = F(y)\|v\|^2$. Thus $\ell(y) = \langle v/\|v\|^2, y \rangle$. Uniqueness follows by taking y equal to the difference of two representing vectors. Its norm equals $\|\ell\|$: Cauchy-Schwarz gives one inequality, and evaluation at the unit vector in the representing vector's direction gives the other. The Cauchy-Schwarz inequality for the Hilbert inner product follows from the positive-form quadratic proof in Section 2.1, so no separate representation result is being imported here.

For fixed g and x , apply this result to $\ell(y) = B_g(x, y)$. There is a unique vector $\Psi(g)x$ with

$$\langle \Psi(g)x, y \rangle = B_g(x, y) = \int g d\mu_{x,y}. \quad (\text{PR38})$$

Linearity in x follows from sesquilinearity and uniqueness of the representing vector. The bound (PR34) gives $\|\Psi(g)\| \leq C\|g\|_\infty$. Linearity in g , positivity for $g \geq 0$, and $\Psi(\bar{g}) = \Psi(g)^*$ follow by testing the corresponding vector equalities against every y , using (PR33)–(PR34) and their conjugation identity. If the original pairings came from Φ , then $\Psi(f) = \Phi(f)$ for continuous f , by (PR30) and uniqueness. If $F_1(x, y) = \langle x, y \rangle$, then $\Psi(1) = I$.

Equation (PR35), followed by the equality between the norm of a vector and the norm of its represented functional, gives

$$\|\Psi(g)x\|^2 \leq C \int |g|^2 d\mu_x. \quad (\text{PR39})$$

Consequently uniformly bounded pointwise convergence of a sequence of Borel functions implies strong operator convergence on each vector, by (PR19): $\|\Psi(g_k)x - \Psi(g)x\|^2 \leq C \int |g_k - g|^2 d\mu_x \rightarrow 0$. Weak convergence follows as well. All convergence assertions here concern sequences.

The constructed map is the unique linear extension of its continuous-function values that preserves uniformly bounded pointwise sequential convergence in every operator pairing. To prove this, let $\widehat{\Psi}$ be another extension with that convergence property, and fix x, y . The continuous approximants to 1_U in Section 1.8 show equality of the two pairings on indicators of open sets. The Borel sets on whose indicators the two pairings agree form a lambda-system: they include K , complementation uses the common value on 1, and a disjoint countable union follows from linearity for partial sums of indicators followed by the assumed sequential convergence. The proved pi-system argument extends equality to all Borel indicators. Linearity gives equality on simple functions, and uniform simple approximation, which is in particular uniformly bounded pointwise sequential approximation, gives equality for every bounded Borel function. This is true for every x, y , so the operators agree.

Multiplicativity is a further property when the original continuous map is a homomorphism; no multiplicativity follows from positivity alone, and none is assumed in the construction above. All representation, measure, polarization, variation, bounded-Borel pairing, regularity, uniqueness, and convergence assertions used in (PR38)–(PR39) have been proved here.

3. The full calculus of a bounded positive contraction

3.1 Foundations and exact statement

Let H be a complex Hilbert space and let $S \in \mathcal{B}(H)$ satisfy

$$S = S^*, \quad (Sx, x) \geq 0, \quad \|S\| \leq 1, \quad \ker S = \{0\}. \quad (\text{BS1})$$

Set $K = [0, 1]$. We prove that there is a unique strongly countably additive orthogonal projection-valued measure E on the Borel subsets of K , with $E(K) = I$, such that

$$S = \int_K t dE(t). \quad (\text{BS2})$$

For every bounded Borel function $f : K \rightarrow \mathbb{C}$ the integral is a bounded operator. For every Borel function finite at each point, including unbounded functions, its closed densely defined integral has exactly the domain

$$D(f(S)) = \left\{ x \in H : \int_K |f|^2 d\mu_x < \infty \right\}, \quad \mu_x(B) = (E(B)x, x), \quad \|f(S)x\|^2 = \int_K |f|^2 d\mu_x. \quad (\text{BS3})$$

We prove all adjoint and product domains and $E(\{0\}) = 0$. No claim that $S \geq cI$ for a positive c is made.

The scalar interval measure theorem, integration, convergence and finite complex-measure uniqueness are proved in Section 1. The representation of bounded Hilbert functionals is proved in Section 2. The remaining foundations are the Hilbert-space axioms, completeness of bounded operators in operator norm, and the elementary algebra and compactness used below.

Every finite positive Borel measure ν on the interval is regular: apply the already proved representation theorem to $f \mapsto \int f d\nu$; its regular representing measure agrees with ν by Section 1.8, whose uniqueness proof does not assume regularity of the competing measure. If μ is a finite complex measure of finite variation and f is bounded Borel, the set function $f\mu : B \mapsto \int_B f d\mu$ is countably additive. Indeed the finite partial-union indicators converge boundedly pointwise, and dominated convergence for $|\mu|$ passes the integrals to the union. Finite partitions give

$$|f\mu|(B) \leq \int_B |f| d|\mu| \leq \|f\|_\infty |\mu|(B).$$

The finite positive measure $|\mu|$ is regular by the preceding argument. Given a Borel B , choose compact $F \subset B \subset U$ with U open and $|\mu|(U \setminus F)$ arbitrarily small. The displayed bound then gives arbitrarily small $|f\mu|(U \setminus F)$. Thus $f\mu$ is regular as well. This proves the weighted-measure fact used in the multiplicativity argument.

3.2 Elementary positivity facts and Bernstein operator weights

If B is a bounded positive self-adjoint operator, the scalar nonnegativity of $(B(x + zy), x + zy)$, for all $z \in \mathbb{C}$, proves

$$|(Bx, y)|^2 \leq (Bx, x)(By, y). \quad (\text{BS4})$$

Indeed, if $(By, y) > 0$, choose the phase and modulus of z to minimize that quadratic expression. If $(By, y) = 0$, any nonzero cross term would make the expression negative for a suitable phase and sufficiently large modulus; thus the cross term is zero. This proves (BS4) in both cases.

Write $\beta = \sup_{\|x\|=1} (Bx, x)$. Taking the supremum over unit y in (BS4) gives $\|Bx\|^2 \leq \beta(Bx, x)$. Therefore $\|B\| \leq \beta$, and the opposite inequality follows from Cauchy–Schwarz. We have proved

$$\|B\| = \beta, \quad \|Bx\|^2 \leq \|B\|(Bx, x). \quad (\text{BS5})$$

Editorial zero-space case. The unit-sphere supremum immediately above is used when $H \neq \{0\}$. If $H = \{0\}$, its unit sphere is empty and that real supremum does not define β . In this case take $\beta = 0$ separately: the only operator is zero, so both equalities in (BS5) hold with both sides zero. Every later vector, projection and integral on this space is zero. In particular the essential operator norm is zero, using the infimum over nonnegative bounds, and the unit-vector arguments for a nonzero projection apply only when such a projection exists. No assumption of a nonzero Hilbert space is added to the theorem.

Apply this to S . The assumptions imply that $I - S$ is positive. Moreover

$$((S - S^2)x, x) = (Sx, x) - \|Sx\|^2 \geq 0. \quad (\text{BS6})$$

Consequently all four operators $I, S, I - S, S(I - S)$ are positive.

For integers $n \geq 1$ and $0 \leq k \leq n$, define

$$W_{n,k} = \binom{n}{k} S^k (I - S)^{n-k}. \quad (\text{BS7})$$

These operators are positive, as can be proved without taking an operator square root. Write $k = 2a + \varepsilon$ and $n - k = 2b + \delta$, where $\varepsilon, \delta \in \{0, 1\}$, and put $Q = S^a (I - S)^b$. All factors are polynomials in the self-adjoint S , so they commute, and $Q = Q^*$. Then

$$W_{n,k} = \binom{n}{k} Q S^\varepsilon (I - S)^\delta Q.$$

The middle factor is one of the four positive operators just proved positive. For each x , the quadratic form is its nonnegative value at Qx , multiplied by $\binom{n}{k}$. This proves positivity. The binomial formula for the commuting operators $S, I - S$ gives

$$\sum_{k=0}^n W_{n,k} = I. \quad (\text{BS8})$$

For a function $f : K \rightarrow \mathbb{C}$, its scalar Bernstein polynomial and corresponding operator are

$$B_n f(t) = \sum_{k=0}^n f(k/n) \binom{n}{k} t^k (1-t)^{n-k}, \quad (B_n f)(S) = \sum_{k=0}^n f(k/n) W_{n,k}. \quad (\text{BS9})$$

If f is real and $a \leq f \leq b$, then (BS7)–(BS8) imply

$$aI \leq (B_n f)(S) \leq bI. \quad (\text{BS10})$$

We next need convergence for a fixed polynomial in coefficient norm, rather than assuming that uniform convergence already implies operator convergence. For a nonnegative integer d , let $c_{d,j}$ be the number of partitions of a d -element set into j nonempty unlabeled subsets. Counting maps from that set to a k -element set by their fibers gives

$$k^d = \sum_{j=0}^d c_{d,j} (k)_j, \quad (k)_j = k(k-1) \cdots (k-j+1), \quad c_{d,d} = 1. \quad (\text{BS11})$$

For $d = 0$, use $c_{0,0} = 1$. Directly canceling factorials in the binomial sum gives

$$\sum_{k=0}^n (k)_j \binom{n}{k} t^k (1-t)^{n-k} = (n)_j t^j.$$

Terms with $k < j$ are zero; after substituting $k = j + \ell$, the remaining factor is the binomial expansion of $(t + (1-t))^{n-j}$. Thus, for fixed d and $n \geq d$,

$$B_n(t^d) = \sum_{j=0}^d c_{d,j} \frac{(n)_j}{n^d} t^j. \quad (\text{BS12})$$

The coefficient at $j = d$ tends to one, and every coefficient with $j < d$ tends to zero. By linearity, $B_n p \rightarrow p$ in coefficient norm for every fixed polynomial p . Since $\|S^j\| \leq 1$, the operator-norm difference between the two evaluated polynomials is at most the sum of the moduli of their coefficient differences. Hence

$$(B_n p)(S) \longrightarrow p(S) \quad \text{in operator norm.} \quad (\text{BS13})$$

If a real polynomial q is nonnegative on K , (BS10) and (BS13) show that $q(S)$ is positive. Positivity is preserved in operator-norm limits because each scalar quadratic form converges.

For a complex polynomial p , let $M = \max_{t \in K} |p(t)|$. The real polynomial $q(t) = M^2 - \overline{p(t)}p(t)$, interpreted on the real variable t , is nonnegative on K . Polynomial algebra and $S = S^*$ give

$$M^2 I - p(S)^* p(S) = q(S) \geq 0.$$

Taking quadratic forms proves the exact bound

$$\|p(S)\| \leq \max_{0 \leq t \leq 1} |p(t)|. \quad (\text{BS14})$$

No spectral inclusion, spectral radius formula, compactness, or spectral theorem has been used.

3.3 Continuous functional calculus

For completeness, the Bernstein polynomials uniformly approximate every continuous complex function on K . The nonnegative scalar weights in (BS9) have total mass one, mean t , and squared displacement

$$\sum_{k=0}^n (k/n - t)^2 \binom{n}{k} t^k (1-t)^{n-k} = \frac{t(1-t)}{n} \leq \frac{1}{4n}. \quad (\text{BS15})$$

The mean and second moment follow from the preceding factorial identity at $j = 1, 2$, using $k^2 = (k)_2 + k$. For $\eta > 0$, let $\omega_f(\eta) = \sup_{|s-t| \leq \eta} |f(s) - f(t)|$. Separating indices with $|k/n - t| \leq \eta$ and using (BS15) for the rest yields

$$\|B_n f - f\|_\infty \leq \omega_f(\eta) + \frac{2\|f\|_\infty}{4n\eta^2}. \quad (\text{BS16})$$

Uniform continuity makes the first term small by choosing η , and then the second is made small by choosing n . Therefore complex polynomials are uniformly dense in $C(K)$.

For $f \in C(K)$, take any polynomials $p_n \rightarrow f$ uniformly and define

$$\Phi(f) = \lim_n p_n(S) \quad \text{in operator norm.} \quad (\text{BS17})$$

Inequality (BS14) makes the sequence Cauchy and proves that the limit is independent of the sequence. It also gives $\|\Phi(f)\| \leq \|f\|_\infty$. The polynomial identities pass to these limits: for example, if $p_n \rightarrow f$ and $q_n \rightarrow g$ uniformly, then $p_n q_n \rightarrow fg$, while boundedness of the approximating sequences and continuity of operator multiplication give $p_n(S)q_n(S) \rightarrow \Phi(f)\Phi(g)$. Therefore

$$\Phi(1) = I, \quad \Phi(t) = S, \quad \Phi(fg) = \Phi(f)\Phi(g), \quad \Phi(\bar{f}) = \Phi(f)^*. \quad (\text{BS18})$$

Linearity is proved in the same way by polynomial sums and scalar multiples. If $f \geq 0$ is continuous, $h = \sqrt{f}$ is continuous and (BS18) gives $\Phi(f) = \Phi(h)^* \Phi(h) \geq 0$. Alternatively positivity follows from the Bernstein approximants. This proves all continuous-calculus facts needed below.

3.4 Scalar measures with the correct complex convention

For each $x \in H$, the functional $L_x(f) = (\Phi(f)x, x)$ is positive and bounded on $C(K)$. The scalar interval measure theorem gives a unique finite positive regular measure μ_x such that

$$(\Phi(f)x, x) = \int_K f d\mu_x, \quad \mu_x(K) = \|x\|^2. \quad (\text{BS19})$$

Define a finite complex measure for $x, y \in H$ by

$$\mu_{x,y} = \frac{1}{4} \sum_{k=0}^3 i^k \mu_{x+i^k y}. \quad (\text{BS20})$$

To verify the sign, if f is real continuous, the form $(\Phi(f)x, y)$ is Hermitian, linear in x and conjugate-linear in y . Expansion gives

$$(\Phi(f)(x + i^k y), x + i^k y) = L_x(f) + L_y(f) + i^{-k}(\Phi(f)x, y) + i^k(\Phi(f)y, x).$$

Multiplying by i^k , summing, and dividing by four leaves exactly $(\Phi(f)x, y)$. Splitting a complex continuous f into its real and imaginary parts consequently proves

$$(\Phi(f)x, y) = \int_K f d\mu_{x,y} \quad (f \in C(K)). \quad (\text{BS21})$$

Uniqueness of finite complex measures from their continuous integrals now proves that $(x, y) \mapsto \mu_{x,y}$ is linear in its first argument, conjugate-linear in its second, and satisfies $\mu_{y,x} = \overline{\mu_{x,y}}$ and $\mu_{x,x} = \mu_x$. For example, the measures representing $\mu_{\alpha x + \beta z, y}$ and $\alpha \mu_{x,y} + \beta \mu_{z,y}$ have equal integrals against every continuous f by (BS21), hence are equal. The other identities follow by the identical uniqueness argument.

For each Borel B , the scalar form $(x, y) \mapsto \mu_{x,y}(B)$ is positive on the diagonal. The same quadratic argument as in (BS4) proves

$$|\mu_{x,y}(B)|^2 \leq \mu_x(B) \mu_y(B). \quad (\text{BS22})$$

Taking a finite measurable partition $B = \bigcup_j B_j$ and applying scalar Cauchy–Schwarz gives

$$\sum_j |\mu_{x,y}(B_j)| \leq \left(\sum_j \mu_x(B_j) \right)^{1/2} \left(\sum_j \mu_y(B_j) \right)^{1/2}.$$

The supremum over finite partitions, which defines total variation, yields

$$|\mu_{x,y}|(B) \leq \mu_x(B)^{1/2} \mu_y(B)^{1/2}, \quad |\mu_{x,y}|(K) \leq \|x\| \|y\|. \quad (\text{BS23})$$

More generally, for nonnegative Borel h ,

$$\int_K h d|\mu_{x,y}| \leq \left(\int_K h^2 d\mu_x \right)^{1/2} \|y\|. \quad (\text{BS24})$$

For a nonnegative simple $h = \sum_j c_j 1_{B_j}$ on disjoint sets, use (BS23) on each set and Cauchy–Schwarz for the finite sum. Increasing simple approximation proves the displayed inequality in general, with infinity permitted. These estimates will control every unbounded cross integral below. **Editorial zero-vector case in the unbounded estimate.** If $y = 0$, (BS23) gives $|\mu_{x,0}|(K) = 0$, so the left side of (BS24) is zero for every nonnegative Borel h , including functions with infinite squared integral against μ_x . Read the right side in this case as zero separately, rather than evaluating an undefined product of infinity and zero. If $y \neq 0$, its norm is positive and the extended inequality follows directly from the stated increasing simple approximation; an infinite right side is permitted. Thus every cross-integral estimate keeps its exact zero case.

3.5 Bounded Borel operators and direct multiplicativity

For bounded Borel f , the sesquilinear form

$$b_f(x, y) = \int_K f d\mu_{x,y}$$

has modulus at most $\|f\|_\infty \|x\| \|y\|$ by (BS23). Hilbert representation gives a unique vector $\Psi(f)x$ such that

$$(\Psi(f)x, y) = \int_K f d\mu_{x,y} \quad (y \in H). \quad (\text{BS25})$$

The form is linear in x , so uniqueness of the representing vector gives linearity of $\Psi(f)$. Its norm is at most $\|f\|_\infty$, so it is bounded. Representation is being used here for a conjugate-linear functional in y : if $F(y)$ is bounded and conjugate-linear, apply the ordinary linear-functional representation to $\overline{F(y)}$, giving $F(y) = (v, y)$. This accounts for the inner-product convention exactly.

The defining scalar integrals give linearity in f , $\Psi(1) = I$, and $\Psi(f) = \Phi(f)$ for continuous f . They also give

$$\Psi(f)^* = \Psi(\overline{f}), \quad \Psi(f) \geq 0 \text{ if } f \geq 0. \quad (\text{BS26})$$

For the adjoint formula, use $\mu_{y,x} = \overline{\mu_{x,y}}$: the conjugate of $\int \bar{f} d\mu_{y,x}$ is $\int f d\mu_{x,y}$.

We prove multiplicativity rather than invoking a Borel extension theorem. First let g be continuous. For every continuous h ,

$$\int_K h d\mu_{\Phi(g)x,y} = (\Phi(h)\Phi(g)x, y) = (\Phi(hg)x, y) = \int_K hg d\mu_{x,y}.$$

The uniqueness of scalar complex measures implies

$$\mu_{\Phi(g)x,y} = g \mu_{x,y}. \quad (\text{BS27})$$

Hence, for bounded Borel f , (BS25) gives

$$\Psi(f)\Phi(g) = \Psi(fg). \quad (\text{BS28})$$

Taking adjoints in (BS28), using (BS18) and (BS26), and then replacing f, g by their conjugates gives the opposite order

$$\Phi(g)\Psi(f) = \Psi(gf). \quad (\text{BS29})$$

Now keep f bounded Borel and test against any continuous h :

$$\int_K h d\mu_{\Psi(f)x,y} = (\Phi(h)\Psi(f)x, y) = (\Psi(hf)x, y) = \int_K hf d\mu_{x,y}.$$

The measure $f\mu_{x,y}$ is finite and regular, as specified in Section 1. Scalar uniqueness therefore proves the decisive identity

$$\mu_{\Psi(f)x,y} = f \mu_{x,y}. \quad (\text{BS30})$$

For any second bounded Borel g , substitute (BS30) in (BS25):

$$(\Psi(g)\Psi(f)x, y) = \int_K g d\mu_{\Psi(f)x,y} = \int_K gf d\mu_{x,y} = (\Psi(gf)x, y).$$

Since this holds for every x, y ,

$$\Psi(g)\Psi(f) = \Psi(gf). \quad (\text{BS31})$$

Together with (BS26), this proves the bounded Borel unital star algebra homomorphism without any Hilbert-space separability assumption. In particular

$$\|\Psi(f)x\|^2 = (\Psi(|f|^2)x, x) = \int_K |f|^2 d\mu_x. \quad (\text{BS32})$$

If $f_n \rightarrow f$ pointwise and $\sup_n \|f_n\|_\infty < \infty$, then dominated convergence in (BS32) applied to $f_n - f$ gives

$$\Psi(f_n)x \rightarrow \Psi(f)x \quad (x \in H). \quad (\text{BS33})$$

Thus bounded pointwise convergence produces strong operator convergence. Only a scalar measure associated with the particular vector is used; there is no countable enumeration of H .

3.6 Projection-valued measure, exact bounded norms, and uniqueness

For a Borel set $B \subset K$, define

$$E(B) = \Psi(1_B). \quad (\text{BS34})$$

Equations (BS26) and (BS31) give

$$E(B)^* = E(B), \quad E(B)^2 = E(B), \quad E(B)E(C) = E(B \cap C), \quad E(K) = I, \quad E(\emptyset) = 0. \quad (\text{BS35})$$

For disjoint Borel sets B_j , the indicator functions of their finite unions converge boundedly pointwise to the indicator of their union. Linearity and (BS33) imply

$$E\left(\bigcup_{j \geq 1} B_j\right)x = \lim_{N \rightarrow \infty} \sum_{j=1}^N E(B_j)x \quad (x \in H). \quad (\text{BS36})$$

More explicitly, the squared norm of the difference is $\mu_x((\bigcup_j B_j) \setminus (\bigcup_{j \leq N} B_j))$, which tends to zero by finite-measure continuity from above. This proves strong countable additivity. Countable additivity in operator norm is neither used nor claimed.

Taking $f = 1_B$ in (BS25) confirms

$$(E(B)x, y) = \mu_{x,y}(B), \quad \mu_x(B) = \|E(B)x\|^2. \quad (\text{BS37})$$

For a Borel set N , the following are equivalent: $E(N) = 0$; $\mu_x(N) = 0$ for every $x \in H$; and $E(N)x = 0$ for every $x \in H$. We call such a set E -null. For bounded Borel f , its exact norm is

$$\|\Psi(f)\| = \inf\{c \geq 0 : E(\{|f| > c\}) = 0\}. \quad (\text{BS38})$$

For any c in this set, (BS32) gives $\|\Psi(f)x\| \leq c\|x\|$. Conversely, if $E(\{|f| > c\}) \neq 0$, choose a unit vector in the range of that projection. Its scalar measure is supported in $\{|f| > c\}$, by (BS35)–(BS37), and has mass one. Thus its integral of $|f|^2$ is strictly greater than c^2 ; indeed $|f|^2 - c^2$ is positive at every point of a set of full scalar measure, and a nonnegative function with zero integral vanishes almost everywhere. Therefore $\|\Psi(f)\| > c$. These implications prove (BS38), including the zero-space case. A sup norm over all of $[0, 1]$ need not equal the operator norm if part of that interval is E -null.

For simple functions, $\Psi(\sum_j c_j 1_{B_j}) = \sum_j c_j E(B_j)$. Every bounded complex Borel function has uniformly convergent finite-range Borel approximations, obtained, for example, by dividing its bounded real and imaginary ranges into intervals of length $1/n$. Contractivity then identifies $\Psi(f)$ with its bounded spectral integral. In particular $\Psi(t) = \Phi(t) = S$, proving (BS2).

To prove uniqueness, suppose F is another orthogonal PVM on K , with $F(K) = I$, strong countable additivity, and $\int_K t dF(t) = S$. The integral of a simple function on disjoint sets is defined as above. Orthogonality gives

$$\left\| \sum_j c_j F(B_j)x \right\|^2 = \sum_j |c_j|^2 \|F(B_j)x\|^2 \leq \max_j |c_j|^2 \|x\|^2.$$

Thus uniform simple approximation defines its bounded integrals and preserves sums, products, and conjugate adjoints, first checked on common finite partitions and then passed to norm limits. Consequently its polynomial integrals are $p(S)$, and its continuous integrals are $\Phi(f)$ by uniform polynomial approximation. For each x , the positive finite measure $(F(\cdot)x, x)$ therefore has exactly the continuous integrals in (BS19). Scalar uniqueness gives $(F(B)x, x) = \mu_x(B) = (E(B)x, x)$ for all B, x . Polarization with the convention in (BS20) gives $(F(B)x, y) = (E(B)x, y)$ for all x, y , so $F(B) = E(B)$. This proves the unique PVM assertion.

3.7 Unbounded Borel functions and the exact closed domain

Let $f : K \rightarrow \mathbb{C}$ be Borel and finite at every point. Define

$$A_n = \{|f| \leq n\}, \quad P_n = E(A_n), \quad f_n = f1_{A_n}, \quad D_f = \left\{x : \int_K |f|^2 d\mu_x < \infty\right\}. \quad (\text{BS39})$$

The sets A_n increase to K , so $P_n \rightarrow I$ strongly by (BS33). For any $x \in H$, (BS32) gives

$$\|\Psi(f_n)x\|^2 = \int_{A_n} |f|^2 d\mu_x, \quad \|\Psi(f_n - f_m)x\|^2 = \int_{A_n \setminus A_m} |f|^2 d\mu_x \quad (n \geq m). \quad (\text{BS40})$$

The sequence $\Psi(f_n)x$ is Cauchy if and only if $x \in D_f$. Sufficiency follows from the vanishing tails of an integrable nonnegative function. For necessity a Cauchy sequence has bounded norms, so monotone convergence in the first formula gives a finite integral over K . Define

$$T_fx = \lim_n \Psi(f_n)x \quad (x \in D_f). \quad (\text{BS41})$$

The set D_f is a vector space: the pointwise identity of the approximating operators gives $\|\Psi(f_n)(x+y)\|^2 \leq 2\|\Psi(f_n)x\|^2 + 2\|\Psi(f_n)y\|^2$; boundedness of these norms and (BS40) imply $x+y \in D_f$. Scalar multiples are immediate. Limits in (BS41) then give linearity of T_f . Equations (BS40)–(BS41) imply exactly

$$\|T_fx\|^2 = \int_K |f|^2 d\mu_x. \quad (\text{BS42})$$

The domain is dense. For any x , $P_n x \in D_f$ since its scalar measure is $1_{A_n} \mu_x$, and hence its $|f|^2$ moment is at most $n^2 \|x\|^2$. The asserted scalar measure formula follows from (BS35) and (BS37): $\mu_{P_n x}(B) = \|E(B)P_n x\|^2 = \mu_x(B \cap A_n)$. As $P_n x \rightarrow x$, density follows.

For every Borel B , this same argument shows $E(B)D_f \subset D_f$. Passing the bounded commutation identities through the limit (BS41) gives

$$E(B)T_fx = T_f E(B)x \quad (x \in D_f), \quad T_f P_n x = \Psi(f_n)x \quad (x \in H). \quad (\text{BS43})$$

For $x \in D_f$, also $P_n T_fx = \Psi(f_n)x$, by the first formula. Thus $P_n x \rightarrow x$ in the graph norm of T_f .

To prove closedness, let $x_k \in D_f$, $x_k \rightarrow x$, and $T_f x_k \rightarrow y$. For fixed n , boundedness and (BS43) imply

$$\Psi(f_n)x = \lim_k \Psi(f_n)x_k = \lim_k P_n T_f x_k = P_n y.$$

Its squared norm is at most $\|y\|^2$. Monotone convergence in (BS40) gives $x \in D_f$. Letting $n \rightarrow \infty$ in the last display gives $T_fx = y$. This proves closedness with its full domain (BS39), not merely a closed restriction.

For $x \in D_f$ and any $y \in H$, (BS24) shows that f is integrable against $\mu_{x,y}$, and dominated convergence gives

$$(T_fx, y) = \int_K f d\mu_{x,y}, \quad \left| \int_K f d\mu_{x,y} \right| \leq \left(\int_K |f|^2 d\mu_x \right)^{1/2} \|y\|. \quad (\text{BS44})$$

If f is bounded, the construction agrees with $\Psi(f)$; if f is merely E -essentially bounded, (BS42) shows that its domain is all H and it equals a bounded representative with the exact norm (BS38). The operators only depend on f outside E -null sets, because the domain integrals and all approximating scalar integrals ignore those sets.

A function with an infinite value on an E -null Borel set is handled by replacing its value on that set by zero. The result is independent of that replacement. The assertion of dense domain in this theorem concerns functions finite E -almost everywhere; no dense-domain assertion is made for an infinite value on a set with nonzero spectral projection.

The same construction covers the completion of the Borel sigma-algebra by subsets of Borel E -null sets, without selecting a single dominating scalar measure. A countable union of E -null Borel sets is E -null: each μ_x gives that union measure zero, and (BS37) applies. Each completed-measurable finite complex function has a Borel representative outside such a null set. To construct one, approximate its truncated real and imaginary parts by finite-range completed-measurable functions with pointwise errors at most $1/n$. Replace the finitely many completed-measurable level sets of each approximation by Borel representatives. The discrepancies across all approximations are contained in one countable union of Borel E -null sets. The resulting sequence consists of Borel functions and converges to the original function off that union. On the Borel set where the sequence converges, take its limit; give it value zero on the complement. This is a Borel representative. Any two representatives agree off an E -null set, and the preceding domain and integral formulas show that they give the same operator. Thus “measurable” may also be read with this precise completed convention.

3.8 Full adjoint, product, and sum domains

We prove the unbounded adjoint formula in both directions. If $v \in D(T_f^*)$, let $w = T_f^*v$. For arbitrary $h \in H$, $P_n h \in D_f$, and the adjoint identity gives

$$(\Psi(f_n)h, v) = (T_f P_n h, v) = (P_n h, w).$$

Bounded adjoints and self-adjointness of P_n turn this into

$$\Psi(\overline{f_n})v = P_n w. \tag{BS45}$$

Taking norms and using monotone convergence yields $\int_K |f|^2 d\mu_v \leq \|w\|^2$, so $v \in D_{\overline{f}}$. Taking limits in (BS45) gives $T_{\overline{f}}v = w$. Conversely, if $v \in D_{\overline{f}}$ and $u \in D_f$, bounded adjoint identities imply

$$(T_f u, v) = \lim_n (\Psi(f_n)u, v) = \lim_n (u, \Psi(\overline{f_n})v) = (u, T_{\overline{f}}v).$$

Thus

$$T_f^* = T_{\overline{f}}, \quad D(T_f^*) = D_{\overline{f}} = D_f. \tag{BS46}$$

The last equality concerns domain sets, not equality of actions for complex f . Real functions give self-adjoint operators.

For $x \in D_g$, (BS43) and (BS42) prove the scalar change-of-measure formula

$$\mu_{T_g x}(B) = \|E(B)T_g x\|^2 = \|T_g E(B)x\|^2 = \int_B |g|^2 d\mu_x. \tag{BS47}$$

Its last equality uses $\mu_{E(B)x} = 1_B \mu_x$. Equality of the measures then gives, for every nonnegative Borel h ,

$$\int_K h d\mu_{T_g x} = \int_K h |g|^2 d\mu_x, \tag{BS48}$$

first for simple h , then by monotone convergence. Therefore the full domain of the product is

$$D(T_f T_g) = \{x \in D_g : T_g x \in D_f\} = D_g \cap D_{fg}. \tag{BS49}$$

This proves both inclusions, since the second required integral is exactly $\int |fg|^2 d\mu_x$, with no discarded g -moment condition.

To identify the action, put $C_n = \{|f| \leq n, |g| \leq n\}$ and $Q_n = E(C_n)$. Then $Q_n \rightarrow I$ strongly. On $Q_n H$, the functions f, g, fg are all bounded; (BS31), (BS41), and (BS43) consequently give, for every x in the domain in (BS49),

$$Q_n T_f T_g x = \Psi(f 1_{C_n}) \Psi(g 1_{C_n}) x = \Psi(fg 1_{C_n}) x = Q_n T_{fg} x.$$

Letting $n \rightarrow \infty$ proves

$$T_f T_g x = T_{fg} x \quad (x \in D_g \cap D_{fg}). \quad (\text{BS50})$$

The intersection is essential. A complete example is $H = \ell^2(\mathbb{N})$, with $S e_j = j^{-1} e_j$. Direct summation proves that S is positive, self-adjoint, injective, and has norm one. Define $F(B)x$ by keeping precisely the coordinates for which $j^{-1} \in B$. Coordinate multiplication proves the projection identities. For disjoint sets B_k , the squared norm of the countable-additivity remainder is the sum of $|x_j|^2$ over coordinates in their union not in the first N sets; this tends to zero by convergence of $\sum_j |x_j|^2$. Thus F is a PVM. Its integral of t is S , first for uniformly approximating simple functions and then by coordinate limits, so uniqueness identifies it with the constructed E . Set $f(t) = t$ and $g(t) = 1/t$ for $t > 0$, with $g(0) = 0$. The measure of $\{0\}$ is zero for this PVM. The operator T_{fg} is the identity on H , while

$$D(T_f T_g) = D_g = \left\{ x : \sum_{j \geq 1} j^2 |x_j|^2 < \infty \right\}.$$

The vector $x_j = 1/j$ belongs to ℓ^2 and not to this domain. Hence the two operators have the same identity action on a proper subspace but different domains. This explicitly demonstrates why (BS49) cannot lose its intersection.

The product does have the precise closure

$$\overline{T_f T_g} = T_{fg}. \quad (\text{BS51})$$

Indeed (BS50) makes it a restriction of the closed operator T_{fg} . For any $x \in D_{fg}$, the vectors $Q_n x$ above lie in $D_g \cap D_{fg}$, and $Q_n x \rightarrow x$, while $T_{fg} Q_n x = Q_n T_{fg} x \rightarrow T_{fg} x$. Thus their product graphs approach every point of the graph of T_{fg} , proving (BS51).

The sum $T_f + T_g$ is defined on exactly $D_f \cap D_g$. The inequality $|f + g|^2 \leq 2|f|^2 + 2|g|^2$ puts this intersection inside D_{f+g} . The same common cutoffs Q_n , followed by strong limits, prove

$$(T_f + T_g)x = T_{f+g}x \quad (x \in D_f \cap D_g), \quad \overline{T_f + T_g} = T_{f+g}. \quad (\text{BS52})$$

For the closure statement, use $x \in D_{f+g}$, the approximants $Q_n x \in D_f \cap D_g$, and $T_{f+g} Q_n x = Q_n T_{f+g} x$. No equality of the initial sum domain with the full domain of the closed sum is assumed.

Applying (BS49)–(BS50) with $\bar{f}, f$, and using $|f|^2 \leq 1 + |f|^4$, gives

$$\begin{aligned} D(T_f^* T_f) &= D(T_f T_f^*) = \left\{ x : \int_K |f|^4 d\mu_x < \infty \right\}, \\ T_f^* T_f &= T_f T_f^* = T_{|f|^2}. \end{aligned} \quad (\text{BS53})$$

Thus normality and the complete squared-modulus domain follow directly. Recursive powers also retain every intermediate-domain requirement. For an integer $q \geq 1$, a vector is in $D(T_f^q)$ precisely when its successive images through T_f^{q-1} lie in D_f . Induction using (BS49)–(BS50) proves

$$D(T_f^q) = \left\{ x : \int_K |f|^{2q} d\mu_x < \infty \right\}, \quad T_f^q = T_{f^q} \text{ on that domain.} \quad (\text{BS53a})$$

At each step the higher moment implies all lower ones because $|f|^{2k} \leq 1 + |f|^{2q}$ for $0 \leq k \leq q$, and $\mu_x(K) = \|x\|^2 < \infty$. Conversely, the product-domain equality requires the higher moment, so the implication is in both directions. The exact norm is $\|T_f^q x\|^2 = \int_K |f|^{2q} d\mu_x$, by (BS42). The case $q = 0$ is the identity on H .

The kernels are also exact:

$$\ker T_f = E(\{f = 0\})H. \quad (\text{BS54})$$

If $T_f x = 0$, (BS42) makes $\mu_x(\{|f| > 0\}) = 0$, by applying it to the union of sets $\{|f| \geq 1/n\}$. Equation (BS37) then gives $x = E(\{f = 0\})x$. Conversely any vector in that range has zero moment and zero image. In particular a function nonzero E -almost everywhere gives an injective multiplier.

3.9 The point zero and the inverse needed by the application

For $S = \Psi(t)$, equation (BS32) gives

$$\|Sx\|^2 = \int_K t^2 d\mu_x. \quad (\text{BS55})$$

If $Sx = 0$, the measure of $[1/n, 1]$ is zero for every n , since the integral is at least $n^{-2}\mu_x([1/n, 1])$. Hence $\mu_x((0, 1]) = 0$ and $x = E(\{0\})x$. Conversely $SE(\{0\}) = \Psi(t1_{\{0\}}) = 0$. We have proved

$$\ker S = E(\{0\})H. \quad (\text{BS56})$$

The assumed injectivity therefore implies $E(\{0\}) = 0$. This does not exclude spectral mass arbitrarily close to zero.

Define the finite-valued Borel function

$$r_0(t) = \begin{cases} t^{-1}, & 0 < t \leq 1, \\ 0, & t = 0. \end{cases} \quad (\text{BS57})$$

Because the point zero is E -null, $T_{tr_0} = I$ on H . The product-domain theorem gives

$$D(T_{r_0} S) = D(S) \cap D(T_{tr_0}) = H, \quad T_{r_0} S = I \text{ on } H.$$

In the opposite order, it gives

$$D(ST_{r_0}) = D(T_{r_0}), \quad ST_{r_0} = I \text{ on } D(T_{r_0}).$$

The first identity says $SH \subset D(T_{r_0})$, and the second says every $u \in D(T_{r_0})$ equals $S(T_{r_0} u)$, so the reverse inclusion also holds. Therefore

$$D(T_{r_0}) = \text{Ran } S, \quad T_{r_0} = S^{-1} \text{ on that exact domain,} \quad \|S^{-1}u\|^2 = \int_{(0,1]} t^{-2} d\mu_u. \quad (\text{BS58})$$

The range is dense: if $y \perp \text{Ran } S$, then $(Sx, y) = 0$ for all x , so $Sy = 0$ by self-adjointness and hence $y = 0$. This also agrees with the dense-domain conclusion already proved for T_{r_0} . **Editorial extension: keep the kernel instead of assuming it away.** The existence, uniqueness, bounded calculus, unbounded domains, adjoints, products, closures and kernel identity in Sections 3.2–3.8 never use $\ker S = \{0\}$. They therefore hold for the same original bounded positive self-adjoint contraction S when its kernel is nonzero. Only the zero-mass conclusion in Section 3.9 uses injectivity. Here is its full replacement, with the point zero retained. Put

$$N = E(\{0\}), \quad P = I - N = E((0, 1]), \quad R = T_{r_0}, \quad D(R) = \left\{ u : \int_{(0,1]} t^{-2} d\mu_u < \infty \right\}. \quad (\text{SE1})$$

The value $r_0(0) = 0$ remains exactly as in (BS57). The finite-valued Borel construction proves that R is densely defined, closed and self-adjoint, with $\|Ru\|^2 = \int_{(0,1]} t^{-2} d\mu_u$. The kernel identity (BS54) gives $\ker S = NH = \ker R$. The product rule (BS50), with the actual function $tr_0(t) = 1_{(0,1]}(t)$, gives both ordered maps and their entire domains:

$$RS = P \quad \text{on } H, \quad SR = P \quad \text{on } D(R). \quad (\text{SE2})$$

Indeed the first domain is $D(S) \cap D(1_{(0,1]}(S)) = H$; the second is $D(R) \cap D(1_{(0,1]}(S)) = D(R)$. Thus $SH \subset D(R)$, while $NH \subset D(R)$ and $RN = 0$, by their exact scalar measures. Also $Sx \in PH$: the bounded product NS has multiplier $1_{\{0\}}(t)t = 0$, so $NS = 0$. For $u \in D(R)$, (SE2) gives $Pu = SRu \in \text{Ran } S$. Consequently

$$D(R) = \text{Ran } S \oplus NH, \quad \overline{\text{Ran } S} = PH. \quad (\text{SE3})$$

The sum is orthogonal because $\text{Ran } S \subset PH$; it is a statement about the actual domain, and does not assert that $\text{Ran } S$ is closed. For the closure equality, a vector is orthogonal to $\text{Ran } S$ exactly when $(Sx, y) = 0$ for all x , equivalently $Sy = 0$ by self-adjointness. Hence its orthogonal complement is NH , and the closed-subspace decomposition proved in Section 6.1 identifies the closure as PH . On PH , the restriction of S is injective, maps onto precisely $\text{Ran } S$, and has inverse $R|_{\text{Ran } S}$: both compositions follow from (SE2), since P is the identity on this subspace. To check that the inverse has its claimed codomain, the kernel identity and self-adjointness of R give $(Ru, n) = (u, Rn) = 0$ for every $n \in NH$. Its norm is the full integral in (SE1). The maps retain the original S , the kernel projection and both composition domains; no lower bound away from zero is inferred.

For a solved model with both defects present, take $H = \ell^2(\mathbb{N}_0)$, $(Su)_0 = 0$ and $(Su)_j = u_j/j$ for $j \geq 1$. Finite coordinate sums prove positivity, self-adjointness and $\|S\| = 1$. Indicator multiplication at the original coordinates $0, 1, 1/2, 1/3, \dots$ is a strongly countably additive PVM: countable additivity follows by taking the squared coordinate sum and its vanishing tails. Its coordinate integral is S , so uniqueness identifies it with E . Then N is the coordinate-zero projection,

$$D(R) = \left\{ u : \sum_{j=1}^{\infty} j^2 |u_j|^2 < \infty \right\}, \quad (Ru)_0 = 0, \quad (Ru)_j = ju_j \quad (j \geq 1). \quad (\text{SE4})$$

The range of S consists precisely of the vectors in this domain whose coordinate zero is zero. Finite sequences with that coordinate zero are dense in PH , but $u_0 = 0, u_j = 1/j$ belongs to PH and not to the range: its squared sum converges by the telescoping comparison used in Section 5.3, while $\sum_{j \geq 1} j^2 |u_j|^2 = \sum_{j \geq 1} 1$ diverges. This proves that the range need not be closed. At the same time $e_0 \in \ker S$, $Re_0 = 0$, and both products in (SE2) kill e_0 . Replacing either product by I would therefore be false. For the injective inverse constructed in Section 4, $N = 0$, so the original (BS58) and all lower-bounded arguments remain unchanged.

4. The original lower-bounded operator and its sharp cutoff

4.1 The bounded inverse is constructed on the actual domain

Let H be a complex Hilbert space and let $A : D(A) \subset H \rightarrow H$ be densely defined and self-adjoint, with

$$(Au, u) \geq a\|u\|^2 \quad (u \in D(A)). \quad (\text{LB1})$$

Fix exactly a real number $\rho \geq 1 + \max(0, -a)$, as in (DSP40), and write

$$T = A + \rho I, \quad D(T) = D(A), \quad \delta = a + \rho \geq 1. \quad (\text{LB2})$$

For $u \in D(T)$, Cauchy–Schwarz and (LB1) give $\|Tu\| \geq \delta\|u\|$. In particular T is injective. It is closed because A is closed: if $u_n \rightarrow u$ and $Tu_n \rightarrow v$, then $Au_n \rightarrow v - \rho u$, so closedness of A gives $u \in D(A)$ and $Tu = v$.

Its range is closed. Indeed, if Tu_n converges, the displayed estimate makes u_n Cauchy. Its limit and the closed graph of T identify the limit of Tu_n as an element of $\text{Ran } T$. Also $T^* = A^* + \rho I = T$, with domain $D(A)$: subtract the bounded functional $\rho(u, v)$ in the definition of the adjoint to obtain both inclusions of domains and the equality of values. The identity

$$(\text{Ran } T)^\perp = \ker T^* = \{0\}$$

follows directly from that same adjoint definition. Thus the range is both closed and dense, and equals H .

Consequently the inverse

$$S = T^{-1} : H \longrightarrow H, \quad \text{Ran } S = D(A), \quad \ker S = \{0\}, \quad \|S\| \leq \delta^{-1} \leq 1 \quad (\text{LB3})$$

exists everywhere and is bounded. The equalities $TS = I$ on H and $ST = I$ on $D(T)$ follow from inverse bijectivity, with exactly the indicated domains. If $x = Tu$, $y = Tv$, then

$$(Sx, y) = (u, Tv) = (Tu, v) = (x, Sy), \quad (Sx, x) = (u, Tu) \geq 0. \quad (\text{LB4})$$

Every $x, y \in H$ have such representations, so $S = S^* \geq 0$. There is no compactness assertion. The range is dense because $D(A)$ is dense.

4.2 The original real spectral coordinate is recovered

The construction in Section 3 gives its unique projection-valued measure E on $[0, 1]$, with

$$S = \int_{[0,1]} t dE(t), \quad \mu_u(C) = (E(C)u, u), \quad \|g(S)u\|^2 = \int |g(t)|^2 d\mu_u(t) \quad (\text{LB5})$$

on the exact square-integrability domain for a finite measurable g . At a null endpoint a chosen finite value makes no difference. Injectivity implies $E(\{0\}) = 0$: by (LB5), $Su = 0$ is equivalent to $\mu_u((0, 1]) = 0$, hence its kernel is exactly $E(\{0\})H$.

Put $c = \delta^{-1}$. The measure is concentrated on $[0, c]$. To prove this without assuming a spectral support theorem, fix $b > c$. For $u \in E([b, 1])H$, (LB5) and the bounded calculus give

$$b\|u\|^2 \leq (Su, u) \leq \|S\|\|u\|^2 \leq c\|u\|^2.$$

Thus $E([b, 1]) = 0$. The union of these sets for rational $b > c$ is $(c, 1]$, and strong countable additivity proves $E((c, 1]) = 0$. If $c = 1$ this set is empty. Combined with the preceding endpoint calculation, the full mass lies on

$$J = (0, \delta^{-1}], \quad \kappa : J \longrightarrow [a, \infty), \quad \kappa(t) = t^{-1} - \rho, \quad \kappa^{-1}(\lambda) = (\lambda + \rho)^{-1}. \quad (\text{LB6})$$

These are inverse continuous maps, with reversed order. Their exact endpoint relation is $\kappa(\delta^{-1}) = \delta - \rho = a$.

For a Borel set $B \subset \mathbb{R}$, define

$$F(B) = E(\{t \in J : \kappa(t) \in B\}). \quad (\text{LB7})$$

Preimages preserve disjoint unions and intersections. Therefore F is a projection-valued measure, strongly countably additive, with $F(\mathbb{R}) = E(J) = I$, and $F((-\infty, a)) = 0$. Its scalar measures satisfy, first for indicator functions, then for nonnegative simple functions and increasing limits,

$$\int_{\mathbb{R}} h(\lambda) d(F(\lambda)u, u) = \int_J h(t^{-1} - \rho) d\mu_u(t) \quad (\text{LB8})$$

for every nonnegative Borel h , with both sides allowed to be infinite. Linear decomposition gives the same change-of-variable identity for integrable complex h . The notation on the left denotes integration against $B \mapsto (F(B)u, u)$, not a derivative of a scalar distribution function.

4.3 Equality of the closed operators, including their domains

Let $G(t) = t^{-1}$ on J , and give it value zero outside J . Its value at zero is immaterial because that set is E -null. The unbounded calculus proved in Section 3 defines the closed operator $G(S)$.

If $u = Sf$, the measure identity for multiplication by the bounded function t is

$$\mu_{Sf}(B) = \int_B t^2 d\mu_f(t). \quad (\text{LB9})$$

Indeed $E(B)S = SE(B)$, and the squared norm of $SE(B)f$ is the displayed integral. Thus

$$\int_J t^{-2} d\mu_{Sf}(t) = \|f\|^2. \quad (\text{LB10})$$

So $\text{Ran } S \subset D(G(S))$ and the product identity gives $G(S)Sf = f$. Conversely, if $u \in D(G(S))$, put $f = G(S)u$. Bounded multiplication by t gives $Sf = u$, since $tG(t) = 1$ on the full-measure set J . Therefore

$$D(G(S)) = \text{Ran } S = D(A), \quad G(S)u = Tu. \quad (\text{LB11})$$

This proves the actual graph equality, not just an identity on a test core.

On J , $G = \kappa + \rho$. The elementary inequalities

$$|G|^2 \leq 2|\kappa|^2 + 2\rho^2, \quad |\kappa|^2 \leq 2|G|^2 + 2\rho^2 \quad (\text{LB12})$$

show equality of their square-integrability domains, since $\mu_u(J) = \|u\|^2 < \infty$. Bounded addition in the calculus consequently gives

$$D(\kappa(S)) = D(A), \quad \kappa(S)u = G(S)u - \rho u = Au. \quad (\text{LB13})$$

Thus the coordinate operator of F is exactly the given A .

4.4 Every measurable multiplier and every recursive power

For any finite-valued complex Borel m on $\mathbb{R}$, define $m(A) = (m \circ \kappa)(S)$, extending the composite arbitrarily across the E -null complement of J . Equations (LB5) and (LB8) prove

$$\begin{aligned} D(m(A)) &= \left\{ u \in H : \int_{\mathbb{R}} |m(\lambda)|^2 d(F(\lambda)u, u) < \infty \right\}, \\ \|m(A)u\|^2 &= \int_{\mathbb{R}} |m(\lambda)|^2 d(F(\lambda)u, u). \end{aligned} \quad (\text{LB14})$$

The bounded measure construction proves density and closedness of this operator, its full adjoint $m(A)^* = \overline{m}(A)$, and the exact product rule

$$\begin{aligned} D(m_1(A)m_2(A)) &= D(m_2(A)) \cap D((m_1m_2)(A)), \\ m_1(A)m_2(A)u &= (m_1m_2)(A)u \quad \text{on this domain.} \end{aligned} \tag{LB15}$$

These statements transfer by (LB8) with both domains retained; no intersection is discarded. For bounded m , (LB14) gives $\|m(A)\| \leq \sup_{\lambda \geq a} |m(\lambda)|$. More exactly the norm is the essential supremum relative to F : the upper bound follows by integration, and if $F(\{|m| > b\}) \neq 0$, a unit vector in this projection range has image norm greater than b . Taking all such b proves equality.

The power in (DSP1) is the recursively defined power of the original operator, not a newly assigned domain. For $q = 0$, it is I on H ; for $q = 1$, (LB13) proves the assertion. If the assertion holds for q , the product rule gives

$$D(A^{q+1}) = \left\{ u : \int |\lambda|^{2q} d(F(\lambda)u, u) < \infty, \quad \int |\lambda|^{2q+2} d(F(\lambda)u, u) < \infty \right\}.$$

The second integral implies the first because $|\lambda|^{2q} \leq 1 + |\lambda|^{2q+2}$. Thus induction proves

$$D(A^q) = \left\{ u : \int |\lambda|^{2q} d(F(\lambda)u, u) < \infty \right\}, \quad A^q = (\lambda \mapsto \lambda^q)(A). \tag{LB16}$$

The opposite recursive convention $A^q A$ has domain $D(A) \cap D((\lambda \mapsto \lambda^{q+1})(A))$ by (LB15). The moment of order $2q + 2$ implies the moment of order two, because $|\lambda|^2 \leq 1 + |\lambda|^{2q+2}$. Thus this domain and action coincide with those of AA^q . Both conventions give (LB16), including the recurrence through Au used in (DSP39).

The same induction applies to $T = A + \rho I$, giving its recursive powers with multiplier $(\lambda + \rho)^q$. The inequalities

$$|\lambda + \rho|^{2q} \leq 2^{2q-1}(|\lambda|^{2q} + \rho^{2q}), \quad |\lambda|^{2q} \leq 2^{2q-1}(|\lambda + \rho|^{2q} + \rho^{2q})$$

for $q \geq 1$ prove $D(T^q) = D(A^q)$, without changing either operator.

Uniqueness is also preserved. Suppose a real projection-valued measure F' represents the same closed A , with its square-integrability multiplier domains. Its lower concentration follows from the operator inequality itself. For

$$C_n = \{\lambda : |\lambda| \leq n, \lambda \leq a - 1/n\},$$

a vector $u \in F'(C_n)H$ belongs to $D(A)$, since the coordinate is bounded on C_n . Therefore

$$a\|u\|^2 \leq (Au, u) = \int_{C_n} \lambda d(F'(\lambda)u, u) \leq (a - 1/n)\|u\|^2.$$

This forces $F'(C_n) = 0$. These sets increase to $(-\infty, a)$, so strong countable additivity gives $F'((-\infty, a)) = 0$. On the full-mass half-line its multiplier $r(\lambda) = (\lambda + \rho)^{-1}$, extended as zero elsewhere, is bounded by δ^{-1} . The product rule shows $(A + \rho I)r(F') = I$ on H and $r(F')(A + \rho I) = I$ on $D(A)$; all domain assertions follow from $(\lambda + \rho)r(\lambda) = 1$. Therefore $r(F') = S$. Push F' forward under r to a measure on $[0, 1]$, assigning zero mass at zero. It represents S , hence equals E by the uniqueness in the bounded construction. The inverse map κ then recovers $F' = F$.

4.5 The precise sharp projector in the boundary application

For the original real parameter λ , put $E_\lambda = F((-\infty, \lambda])$. If $\lambda < a$, this projection is zero. If $\lambda \geq a$, its exact inverse-coordinate description is

$$E_\lambda = E([\lambda + \rho)^{-1}, (a + \rho)^{-1}]. \quad (\text{LB17})$$

Both endpoints are included. In particular the change of coordinate reverses the inequality and keeps any atom at λ .

For $u \in E_\lambda H$, its F -measure is supported in $[a, \lambda]$. All moments of this finite interval are bounded, so $u \in D(A^q) = D(T^q)$ for every integer $q \geq 0$. Write $L = \lambda + \rho \geq 1$. On this interval $1 \leq a + \rho \leq t + \rho \leq L$. Therefore

$$\|T^j u\| \leq L^j \|u\|, \quad \|A^j u\| \leq (L + \rho)^j \|u\| \leq (1 + \rho)^j L^j \|u\|. \quad (\text{LB18})$$

Since $L \geq 1$ and $1 + \rho \geq 1$, summing gives the explicit instance of (DSP40)

$$\sum_{j=0}^q \|A^j u\| \leq (q + 1)(1 + \rho)^q L^q \|u\|. \quad (\text{LB19})$$

This estimate is valid for every $\lambda \geq a$; the later local kernel argument uses $\lambda \geq 1$ to compare L with λ . If the projection is zero the estimate holds as well.

For a bounded Borel m supported in $(-\infty, b]$, bounded multiplicativity gives $m(A) = E_b m(A) E_b$ exactly, including its value at b . For a bounded Borel sequence m_n with a common uniform bound and pointwise convergence to m , (LB14) applied to the differences and dominated convergence give $m_n(A)u \rightarrow m(A)u$ for every $u \in H$.

Finally, the entire proof applies directly to $H = L^2(X, r dx)$ with its original weighted inner product. No unitary change of density, coordinate change, compactness assumption, boundary condition or alteration of the test-domain inclusion (DSP33) enters this spectral step. Every differential-domain and boundary-regularity assertion in Section 10 of Dirichlet realizations, spectral projectors, and local extensions consequently keeps its existing separate proof. **Editorial receiving maps for an original real Hilbert space.** The preceding sections explicitly use a complex Hilbert space. They also give a spectral theorem for a given real Hilbert space without replacing its inner product or domain. Keep the original real space $H_{\mathbb{R}}$, its inner product $[x, y]$, and construct the complete pair space from (BC8) in Banach foundations:

$$H_{\mathbb{C}} = H_{\mathbb{R}} \times H_{\mathbb{R}}, \quad (a + ib)(x, y) = (ax - by, bx + ay), \quad ((x, y), (u, v)) = [x, u] + [y, v] + i([y, u] - [x, v]). \quad (\text{SE5})$$

Its norm square is $\|x\|^2 + \|y\|^2$. The original injection $\iota x = (x, 0)$ is isometric; the component maps are real-linear contractions. The conjugation $C(x, y) = (x, -y)$ satisfies $C^2 = I$, $(Cz, Cw) = \overline{(z, w)}$, and has fixed space exactly $\iota H_{\mathbb{R}}$.

For any densely defined real-linear operator $L : D(L) \subset H_{\mathbb{R}} \rightarrow K_{\mathbb{R}}$, define $L_{\mathbb{C}}(x, y) = (Lx, Ly)$ on precisely $D(L) \times D(L)$. This domain is dense by component approximation. Component norm convergence proves that $L_{\mathbb{C}}$ is closed if and only if L is closed. Define the real adjoint by $[Lx, u]_K = [x, L^*u]_H$ for every $x \in D(L)$, with its full bounded-pairing domain. Then

$$D(L_{\mathbb{C}}^*) = D(L^*) \times D(L^*), \quad L_{\mathbb{C}}^*(u, v) = (L^*u, L^*v). \quad (\text{SE6})$$

To prove the first inclusion, let (u, v) be in the complex adjoint domain with adjoint value (p, q) . Testing on $(x, 0)$ gives $[Lx, u] - i[Lx, v] = [x, p] - i[x, q]$. Equality of real and imaginary parts is exactly the two real adjoint identities, so $u, v \in D(L^*)$, $p = L^*u$, $q = L^*v$. Conversely substitute these two real

identities into every one of the four terms of (SE5) for an arbitrary (x, y) to obtain the full complex adjoint identity. This proves both domains and values. Applying it to L^* is legitimate whenever that operator is densely defined; no such density is assumed for a general L here.

Let the original real $A = A^*$ be densely defined with $[Ax, x] \geq a\|x\|^2$. Formula (SE6) makes $A_{\mathbb{C}}$ self-adjoint on exactly $D(A) \times D(A)$, and expansion gives

$$(A_{\mathbb{C}}(x, y), (x, y)) = [Ax, x] + [Ay, y] + i([Ay, x] - [Ax, y]) = [Ax, x] + [Ay, y] \geq a(\|x\|^2 + \|y\|^2). \quad (\text{SE7})$$

The imaginary terms cancel by the original real symmetry. Use the unchanged $\rho, \delta, T = A + \rho I$ and $D(T) = D(A)$. Section 4 gives the complex inverse of $T_{\mathbb{C}}$. Since $T_{\mathbb{C}}(u, v) = (Tu, Tv)$ is bijective, testing right-hand sides $(x, 0)$ proves that T is bijective on its actual real domain. Its inverse is $S = T^{-1}$, and the complex inverse is exactly $S_{\mathbb{C}}$. Its norm agrees with the real inverse norm: the component sum gives the upper bound, and the isometric injection gives the reverse bound. All original inverse estimates therefore retain the same constants.

The bounded spectral construction, including the noninjective extension (SE1)–(SE3), applies also to the complexification of any real bounded positive contraction. Its PVM commutes with C . Indeed $E'(B) = CE(B)C$ is a complex-linear orthogonal PVM: the two conjugations give linearity, preserve products and adjoints by (SE5), and preserve every strong vector limit. Its coordinate integral is $CS_{\mathbb{C}}C = S_{\mathbb{C}}$, as follows first for real simple approximants and then by their uniform limits. Bounded uniqueness gives $E' = E$. Thus the original real operators

$$E_{\mathbb{R}}(B)x = \text{Re}(E(B)\iota x), \quad E(B)\iota = \iota E_{\mathbb{R}}(B), \quad E(B)(x, y) = (E_{\mathbb{R}}(B)x, E_{\mathbb{R}}(B)y) \quad (\text{SE8})$$

are orthogonal projections and a strongly countably additive real PVM. The first formula is well-defined because commutation makes $E(B)\iota x$ fixed by C . The second gives all real products, pairings and strong limits by the isometric injection; complex-linearity then gives the third. For a pair $z = (x, y)$, its scalar measure is exactly $\mu_z = \mu_x^{\mathbb{R}} + \mu_y^{\mathbb{R}}$: (SE5) and real self-adjointness of the projections cancel both imaginary cross terms. In particular there is no discarded component contribution.

For the actual lower-bounded A , the same original map $\kappa(t) = t^{-1} - \rho$ on $J = (0, \delta^{-1}]$ pushes these real projections to $F_{\mathbb{R}}$. The complex F is its componentwise complexification. For any finite-valued real Borel f on $\mathbb{R}$, bounded truncations preserve the fixed space, so their graph limit does as well. Define its restriction by

$$D(f(A)_{\mathbb{R}}) = \left\{ x : \int |f(\lambda)|^2 d\mu_x^{\mathbb{R}} < \infty \right\}, \quad f(A)\iota x = \iota f(A)_{\mathbb{R}}x. \quad (\text{SE9})$$

The full complex moment domain is the product of two displayed real domains, because the pair moment is their sum. The truncated actions and their graph limits therefore give $f(A) = (f(A)_{\mathbb{R}})_{\mathbb{C}}$ on that entire product domain. Spectral cutoffs show density of the real domain; component limits show closedness. Apply (SE6) to this dense real operator and the proved complex identity $f(A)^* = f(A)$. It gives real self-adjointness with both real adjoint domains equal to (SE9). Products of two real multipliers retain exactly $D(g(A)_{\mathbb{R}}) \cap D((fg)(A)_{\mathbb{R}})$, since (LB15) restricts to this moment domain through the injection. The coordinate operator is the original real A by (LB13); recursive powers, both inverse products, the closed-endpoint projector (LB17), and every constant in (LB18)–(LB19) consequently transfer on their exact original real domains.

For clarity, a complex-valued $f = f_1 + if_2$ need not define an operator from the real space to itself. Its exact receiving map is instead

$$f(A)\iota x = (f_1(A)_{\mathbb{R}}x, f_2(A)_{\mathbb{R}}x), \quad D(f(A)\iota) = D(f_1(A)_{\mathbb{R}}) \cap D(f_2(A)_{\mathbb{R}}), \quad (\text{SE10})$$

and its norm square is the sum of the two squared component norms. The identity $|f|^2 = f_1^2 + f_2^2$ proves both inclusions of the domain equality without removing either term. On the common cutoff $\{|f| \leq n\}$,

bounded scalar linearity proves the component action. These cutoffs converge in the graph norm of each component, by their exact moment integrals, so closedness gives the full displayed action. Thus a complex-valued multiplier keeps its actual codomain as well as its domain. No real scalar action by i is presumed.

5. Two exact models and a solved domain exercise

The same coordinate correspondence can represent continuous spectrum or a sequence of eigenvalues. Here both models use the actual constants $a = -2$, $\rho = 3$, $\delta = 1$, and the cutoff $\lambda = 4$.

5.1 A weighted continuous model

Let

$$H = L^2([0, \infty), e^{-x} dx), \quad D(A) = \left\{ u \in H : \int_0^\infty |x - 2|^2 |u(x)|^2 e^{-x} dx < \infty \right\}, \quad Au(x) = (x - 2)u(x). \quad (\text{SC1})$$

This is densely defined: truncating any $u \in H$ to $[0, n]$ gives domain elements converging in H . Multiplication by the real function $x - 2$ is symmetric there. To identify its full adjoint, let $v \in D(A^*)$ and put $w = A^*v$. Test the adjoint identity with every $u \in H$ supported in $[0, n]$, which belongs to $D(A)$. It gives $1_{[0, n]}w = (x - 2)1_{[0, n]}v$ in H . Letting n increase proves $w = (x - 2)v$ almost everywhere, so $v \in D(A)$. The converse follows by the integrable Hilbert pairing. Thus $A = A^*$ on exactly (SC1), and

$$(Au, u) = \int_0^\infty (x - 2)|u(x)|^2 e^{-x} dx \geq -2\|u\|^2.$$

The integral is finite for $u \in D(A)$ by Cauchy–Schwarz.

Here $T = A + 3I$ multiplies by $x + 1$, and $S = T^{-1}$ multiplies by $(x + 1)^{-1}$, with no density change. Its PVM on $[0, 1]$ is

$$E(B)u(x) = 1_B((x + 1)^{-1})u(x). \quad (\text{SC2})$$

Indicator multiplication proves idempotence, adjoints and intersections; dominated convergence against $|u|^2 e^{-x} dx$ proves strong countable additivity. Its coordinate integral is S , so bounded uniqueness identifies it with the constructed PVM. Therefore the real PVM and sharp projector are exactly

$$F(B)u(x) = 1_B(x - 2)u(x), \quad E_\lambda u(x) = \begin{cases} 1_{[0, \lambda + 2]}(x)u(x), & \lambda \geq -2, \\ 0, & \lambda < -2. \end{cases} \quad (\text{SC3})$$

An eigenvector with eigenvalue b would vanish outside the singleton $\{b + 2\}$, which has measure zero for $e^{-x} dx$; hence there are no nonzero eigenvectors. For $\lambda > -2$, put $c = \lambda + 2 > 0$. The intervals $(c2^{-(j+1)}, c2^{-j})$, $j \geq 0$, are pairwise disjoint subsets of $[0, c]$ with positive weighted measure. Their indicator functions, divided by the square roots of those exact measures, give infinitely many orthonormal vectors in $E_\lambda H$. Thus this projection has infinite rank. A local spectral argument cannot infer finite rank from a finite spectral interval alone.

At the specified cutoff,

$$E_4 u = 1_{[0, 6]} u, \quad E_4 = E([1/7, 1]). \quad (\text{SC4})$$

The example keeps the nonconstant density and the original coordinate x .

5.2 An atomic model that detects the endpoint

On $H = \ell^2(\mathbb{N}_0)$, define

$$D(A) = \left\{ u : \sum_{j=0}^{\infty} |j-2|^2 |u_j|^2 < \infty \right\}, \quad (Au)_j = (j-2)u_j. \quad (\text{SC5})$$

Finite sequences are dense. Testing the adjoint identity on each coordinate vector forces the adjoint value to be $(j-2)v_j$; membership of that sequence in ℓ^2 is exactly the displayed domain. Conversely Cauchy–Schwarz proves the adjoint identity on that domain. Thus this is self-adjoint and lower-bounded by -2 . Its inverse $S = (A + 3I)^{-1}$ multiplies coordinate j by $(j+1)^{-1}$. The coordinate projections give its PVM directly, and

$$E_4 u = (u_0, u_1, \dots, u_6, 0, \dots), \quad E_4 = E([1/7, 1]). \quad (\text{SC6})$$

Both equalities include coordinate $j = 6$. Replacing the lower endpoint $1/7$ by an open endpoint deletes that eigenspace. This proves that the endpoint convention in (LB17) has observable mathematical content, even though the continuous model gives individual endpoints measure zero.

5.3 Exercise and complete solution

For the atomic model, determine the domains and actions of $S(A + 3I)$ and of the multiplier obtained from the product $t \cdot t^{-1} = 1$ on $(0, 1]$. Exhibit a vector that distinguishes them.

The operator $A + 3I$ has domain (SC5), and S is bounded on all of H . Thus $S(A + 3I)$ has exactly domain $D(A)$; its coordinate action there is $u_j \mapsto (j+1)^{-1}(j+1)u_j = u_j$. The product function equals one outside the E -null point zero, so its multiplier is I on all of H .

Take $u_j = (j+1)^{-1}$. Its squared sum is finite: for $k \geq 2$, $k^{-2} \leq [k(k-1)]^{-1} = (k-1)^{-1} - k^{-1}$, whose partial sums are bounded. But for $j \geq 5$, $(j-2)/(j+1) \geq 1/2$, so the sum in (SC5) is infinite. Hence $u \in H \setminus D(A)$. The two identity actions have different domains exactly as (LB15) states. The opposite composition $(A + 3I)S$ is defined on all of H , because multiplication by $(j+1)^{-1}$ places every sequence in $D(A + 3I)$, and its action is I there.

6. Hilbert interfaces supplied by the same construction

The spectral argument also supplies the Hilbert interfaces used by Traces that survive passage to cohomology. We give their arguments here with arbitrary Hilbert dimension, the same linear-first convention, and the exact closed-operator domains. For the orthonormal-basis construction we use the explicitly selected maximality axiom: a partially ordered set whose chains have upper bounds has a maximal element. This is the same foundational choice used in the metric and Hahn–Banach lessons.

6.1 Closed subspaces, arbitrary bases and bounded adjoints

Let M be a closed linear subspace of a complex Hilbert space H . For $x \in H$, set $d = \inf_{m \in M} \|x - m\|$ and choose $m_j \in M$ with $\|x - m_j\| \rightarrow d$. The midpoint is in M , so

$$\|m_j - m_k\|^2 \leq 2\|x - m_j\|^2 + 2\|x - m_k\|^2 - 4d^2 \rightarrow 0. \quad (\text{HF1})$$

Completeness and closedness give a limit $m \in M$ attaining the distance. The minimum of $\|x - m - tz\|^2$, for $z \in M$ and every complex t , gives $(x - m, z) = 0$, by taking real and purely imaginary t . Thus $x = m + (x - m)$ is the unique decomposition in $M \oplus M^\perp$: the intersection is zero, and the preceding calculation supplies existence. Uniqueness proves that $Q_M x = m$ is linear. Orthogonality gives

$$\|x\|^2 = \|Q_M x\|^2 + \|(I - Q_M)x\|^2, \quad Q_M^2 = Q_M = Q_M^*, \quad \|Q_M\| \leq 1. \quad (\text{HF2})$$

The adjoint equality here follows directly from the decomposition, so it does not presuppose the general adjoint theorem.

Order the orthonormal subsets of H by inclusion. The union of a chain is again orthonormal, since any two of its vectors occur together in a member of the chain. Maximality gives an orthonormal family $(e_\alpha)_{\alpha \in I}$. For a finite $F \subset I$, the exact orthogonal decomposition against its finite span gives $\sum_{\alpha \in F} |(x, e_\alpha)|^2 \leq \|x\|^2$. For each positive integer n , the set of indices with $|(x, e_\alpha)| \geq 1/n$ has at most $n^2 \|x\|^2$ elements, by applying this inequality to any finite subset. Therefore the nonzero coefficients of x form a countable set. The bounded partial squared sums have a finite supremum, and their tails tend to zero along an enumeration of that set. The corresponding finite vector sums are Cauchy and converge in H . Their limit has the same coefficient as x at every index. The difference is orthogonal to all e_α ; if it were nonzero, its unit vector could be added to the maximal family. Hence the difference is zero. Every finite set containing a sufficiently long initial segment of the nonzero coefficients has the same small squared tail bound. This proves convergence over all finite subsets, and

$$x = \lim_{F \in I} \sum_{\alpha \in F} (x, e_\alpha) e_\alpha, \quad \|x\|^2 = \sum_{\alpha \in I} |(x, e_\alpha)|^2, \quad (x, y) = \sum_{\alpha \in I} (x, e_\alpha) \overline{(y, e_\alpha)}. \quad (\text{HF3})$$

The last sum is absolutely convergent by finite Cauchy–Schwarz followed by the supremum of the partial sums, and its equality follows from the vector limits. Thus neither separability nor a countable basis was assumed.

For a bounded $L : H \rightarrow K$, the bounded linear functional $x \mapsto (Lx, y)_K$ has, by the Hilbert representation already proved in Section 2, a unique vector $L^*y \in H$ with

$$(Lx, y)_K = (x, L^*y)_H, \quad \|L^*y\| \leq \|L\| \|y\|. \quad (\text{HF4})$$

Uniqueness and the conjugate-linearity of the second argument prove linearity of $y \mapsto L^*y$. Taking the supremum over unit x, y in the pairing gives both $\|L^*\| \leq \|L\|$ and $\|L\| \leq \|L^*\|$. The same identity, tested against all x , gives

$$(\text{Ran } L)^\perp = \ker L^*. \quad (\text{HF5})$$

For an unbounded densely defined L , this range identity still holds with its actual adjoint domain: orthogonality is exactly the bounded zero functional in the definition of L^* . Closedness is not required for this identity.

6.2 Positive square roots on the original spectral coordinate

Let B be a positive self-adjoint operator with its actual domain, and let F_B be its spectral measure constructed in Section 4 with $a = 0, \rho = 1$. The support is $[0, \infty)$. Define the finite Borel function $\sqrt{t}$ there, and zero for $t < 0$. The multiplier theorem gives the positive self-adjoint operator

$$R = (t \mapsto \sqrt{t})(B), \quad D(R) = \left\{ u : \int_{[0, \infty)} t d(F_B(t)u, u) < \infty \right\}. \quad (\text{HF6})$$

Positivity follows from the nonnegative spectral integral of its quadratic form; that integral is finite on $D(R)$ by Cauchy–Schwarz. The product rule retains $D(R) \cap D(B)$ for R^2 . Since $t \leq 1 + t^2$ and the scalar measure has mass $\|u\|^2$, the second domain implies the first. Therefore

$$D(R^2) = D(B), \quad R^2 u = Bu, \quad \|Ru\|^2 = \int t d(F_B(t)u, u). \quad (\text{HF7})$$

Every domain and every factor remains on the original t coordinate.

This positive root is unique even when unbounded. If C is another positive self-adjoint operator with $C^2 = B$ as an equality of operators and domains, its spectral measure F_C is supported in $[0, \infty)$. Push that measure forward by the actual map $s \mapsto s^2$:

$$G(T) = F_C(\{s \geq 0 : s^2 \in T\}). \quad (\text{HF8})$$

The coordinate multiplier of G has precisely the moment domain $\{u : \int s^4 d(F_C(s)u, u) < \infty\} = D(C^2)$ and action C^2 , by (LB16). Thus it represents exactly B , and the uniqueness proved in Section 4 gives $G = F_B$. Squaring is a Borel bijection of $[0, \infty)$ with inverse $\sqrt{t}$. Taking its inverse images in (HF8) recovers F_C from F_B ; the square-root multiplier has precisely the moment domain $\{u : \int s^2 d(F_C(s)u, u) < \infty\} = D(C)$, and action C . Hence $C = R$, including its full domain.

For a bounded positive B , put $M = \|B\|$. Its spectral measure has no mass above M . Indeed for $T_n = \{t : M + 1/n \leq t \leq n\}$, a vector $u \in F_B(T_n)H$ lies in $D(B)$ and satisfies

$$(M + 1/n)\|u\|^2 \leq (Bu, u) \leq M\|u\|^2. \quad (\text{HF9})$$

The projection is zero. The increasing union of T_n is (M, ∞) , so strong countable additivity proves the support assertion. Formula (HF6) consequently has domain all of H , with $\|R\| \leq \sqrt{M}$; (HF7) gives $\|Ru\|^2 = (Bu, u)$. This proves the bounded root theorem from the unchanged B , without an eigenbasis premise or a scalar rescaling of its working expression.

6.3 Closed operators and polar decomposition with both domains proved

Let $S : D(S) \subset H \rightarrow K$ be closed and densely defined. Give $D(S)$ the inner product

$$q(u, v) = (u, v)_H + (Su, Sv)_K. \quad (\text{HF10})$$

Its norm is complete: a q -Cauchy sequence has $u_j \rightarrow u$ in H and $Su_j \rightarrow w$ in K , and closedness gives $u \in D(S)$, $Su = w$. For $f \in H$, Hilbert representation in this complete space produces one $u \in D(S)$ with

$$q(u, v) = (f, v)_H \quad (v \in D(S)). \quad (\text{HF11})$$

The estimate $q(u, u) \leq \|f\|\|u\|$ gives $\|u\| \leq \|f\|$. Conjugating (HF11) gives $(Sv, Su)_K = (v, f - u)_H$ for every $v \in D(S)$. This is exactly the adjoint definition: $Su \in D(S^*)$ and $S^*Su = f - u$. Conversely every $u \in D(S^*S)$ satisfies (HF11) with $f = u + S^*Su$. Thus

$$(I + S^*S) : D(S^*S) \rightarrow H \text{ is a bijection,} \quad T = (I + S^*S)^{-1} : H \rightarrow H, \quad \|T\| \leq 1. \quad (\text{HF12})$$

Injectivity follows from the positive identity $((I + S^*S)u, u) = \|u\|^2 + \|Su\|^2$. For $u = Tf, v = Tg$, (HF11) and symmetry of q give $(Tf, g) = (f, Tg)$ and $(Tf, f) = q(Tf, Tf) \geq 0$. Hence T is bounded positive self-adjoint. If $Tf = 0$, (HF11) says $(f, v) = 0$ for every $v \in D(S)$; density gives $f = 0$. Formula (HF5) now shows that $\text{Ran } T = D(S^*S)$ is dense.

Write $B = S^*S$ on exactly

$$D(B) = \{u \in D(S) : Su \in D(S^*)\}. \quad (\text{HF13})$$

It is symmetric and positive there, since $(Bu, v) = (Su, Sv)$. It is self-adjoint: for $w \in D(B^*)$, choose $v \in D(B)$ with $(I + B)v = (I + B^*)w$, using (HF12). For every $u \in D(B)$, the adjoint identity gives $((I + B)u, w - v) = 0$. Surjectivity in (HF12) gives $w = v \in D(B)$. The reverse adjoint inclusion follows from symmetry, proving equality of domains and actions.

Let $R = B^{1/2}$ be the positive root (HF6). For $u \in D(B)$,

$$\|Su\|^2 = (Bu, u) = \|Ru\|^2. \quad (\text{HF14})$$

We now prove $D(R) = D(S)$, rather than assigning either domain by notation. First $D(B)$ is dense in the graph space $(D(S), q)$. If w is q -orthogonal to its range TH , symmetry and (HF11) give $q(w, Tf) = (w, f)_H = 0$ for every f , hence $w = 0$. Applying the closed-subspace decomposition (HF2) in this graph Hilbert space proves the density. For $u \in D(S)$, choose $u_j \in D(B)$ converging in the S -graph norm. Equation (HF14) applied to $u_j - u_k$ makes Ru_j Cauchy. Closedness of R gives $u \in D(R)$ and $\|Ru\| = \|Su\|$.

Conversely, for $u \in D(R)$, put $u_n = F_B([0, n])u$. Bounded spectral support gives $u_n \in D(B)$. Monotone convergence and the exact norm formula give $u_n \rightarrow u$ and $Ru_n \rightarrow Ru$. Equation (HF14) applied to differences makes Su_n Cauchy. Closedness of S yields $u \in D(S)$, with the same norm equality. We have proved

$$D(S) = D(R), \quad \|Su\| = \|Ru\|, \quad \ker S = \ker R. \quad (\text{HF15})$$

Define U_0 on $\text{Ran } R$ by $U_0(Ru) = Su$. If two inputs Ru agree, their difference has zero R -norm, and (HF15) gives zero S -norm; thus the definition is independent of the preimage. It is linear and isometric. Extend by norm limits to $\overline{\text{Ran } R}$, with range $\overline{\text{Ran } S}$. Its range equals that closure because the range of an isometry from a complete space is closed, and contains the dense original range. Extend it by zero on the orthogonal complement. The densely defined self-adjoint R obeys (HF5), so

$$\overline{\text{Ran } R} = (\ker R)^\perp, \quad S = UR \text{ on exactly } D(S) = D(R). \quad (\text{HF16})$$

This is the polar decomposition, with initial space $(\ker S)^\perp$ and final space $\overline{\text{Ran } S}$. The isometry on that initial space is uniquely determined by the values $U(Ru) = Su$ on its dense subset; zero on the orthogonal complement determines the full U . No closed-range assumption enters the proof.

For any positive self-adjoint B , the projections needed for finite spectral bands are also exact. Put $P_n = F_B([1/n, n])$. On their ranges,

$$P_n H \subset D(B), \quad n^{-1}\|x\| \leq \|Bx\| \leq n\|x\|, \quad P_n \rightarrow I - F_B(\{0\}) = Q_{(\ker B)^\perp} \text{ strongly.} \quad (\text{HF17})$$

The first two claims are respectively the finite moment and the squared multiplier norm on this interval. The intervals increase to $(0, \infty)$; strong countable additivity gives the limit, and the zero-kernel identity (BS54), transferred by (LB8), identifies its range. Multiplicativity gives invariance of $D(B)$ and $BP_n u = P_n Bu$ there. The bounded multiplier $t^{-1}1_{[1/n, n]}(t)$ maps $P_n H$ to itself. Both its products with B have the full domain $P_n H$, by (LB15), and equal the identity on that space. Its norm is at most n . Thus B restricted to this actual spectral range is the claimed bounded bijection with its bounded inverse.

6.4 The Hilbert tensor completion

For complex Hilbert spaces H, K , equip their algebraic tensor product with the sesquilinear form determined by

$$(x \otimes y, x' \otimes y') = (x, x')_H (y, y')_K. \quad (\text{HF18})$$

To check positivity and independence of presentations, take finite orthonormal bases of the spans of all first and second factors in two proposed presentations. Bilinearity expresses either tensor in the same finite basis $e_a \otimes f_b$; its coefficients are recovered by the algebraic linear functionals $x \mapsto (x, e_a)$ and $y \mapsto (y, f_b)$ on that finite tensor space. The universal property of the algebraic tensor product therefore makes the recovered coefficients presentation-independent. Formula (HF18) gives their squared norm exactly as $\sum_{a,b} |c_{ab}|^2$, zero only for the zero tensor. This also proves Cauchy–Schwarz and the triangle inequality by finite coordinate calculation.

Its completion can be constructed as norm-Cauchy sequences modulo sequences tending to zero. Addition and scalar multiplication act componentwise, and the inner product is the limit of the inner products

of representatives; Cauchy–Schwarz makes these limits exist and independent of representatives. The original tensor space embeds isometrically by constant sequences and is dense: the constant sequences at successive terms of any representative approach its class. For a Cauchy sequence w_j of classes, choose algebraic tensors v_j with $\|w_j - v_j\| < 2^{-j}$. Then $\|v_j - v_k\| \leq 2^{-j} + \|w_j - w_k\| + 2^{-k}$, so (v_j) is Cauchy. Its class w satisfies $\|v_j - w\| = \lim_{k \rightarrow \infty} \|v_j - v_k\| \rightarrow 0$. The triangle inequality gives $\|w_j - w\| \leq 2^{-j} + \|v_j - w\| \rightarrow 0$, proving completeness with the full constructed norm.

If (e_α) and (f_β) are any orthonormal bases, the tensors $e_\alpha \otimes f_\beta$ are orthonormal by (HF18). For each elementary tensor, finite-subset approximants $x_F \rightarrow x, y_G \rightarrow y$ from (HF3) give

$$\|x \otimes y - x_F \otimes y_G\| \leq \|x - x_F\| \|y\| + \|x_F\| \|y - y_G\| \rightarrow 0. \quad (\text{HF19})$$

The exact elementary norm used here is $\|x \otimes y\| = \|x\| \|y\|$, also from (HF18). Finite sums of elementary tensors are dense by definition. Therefore the displayed orthonormal tensors have dense span and form an orthonormal basis of the Hilbert completion. The finite-subset convention preserves arbitrary index types throughout.

6.5 Complete direct sums, the adjoint norm identity and geometric inverses

These arguments supply the precise Hilbert inputs for When a moving symbol scale controls an operator and From local energy to global divergence equations. The spaces and operator products retain their original norms and order.

Let $H_j, j \geq 1$, be complex Hilbert spaces, which need not be copies of one space. Define

$$\mathcal{H} = \left\{ u = (u_j) : u_j \in H_j, \sup_{N \geq 1} \sum_{j=1}^N \|u_j\|_{H_j}^2 < \infty \right\}, \quad (u, v)_{\mathcal{H}} = \sum_{j=1}^{\infty} (u_j, v_j)_{H_j}. \quad (\text{HF20})$$

Finite Cauchy–Schwarz bounds the sum of the absolute values in the second formula by $\|u\|_{\mathcal{H}} \|v\|_{\mathcal{H}}$. It therefore converges absolutely. Coordinate addition and scalar multiplication preserve the defining finite-sum bound; sesquilinearity passes through the convergent sums. The squared norm is exactly the sum of the squared coordinate norms, so it is zero only when every coordinate is zero. Finite-coordinate Cauchy–Schwarz and its limit give the triangle inequality.

To prove completeness, let $u^{(k)}$ be Cauchy in this norm. Each coordinate is Cauchy because $\|u_j^{(k)} - u_j^{(l)}\| \leq \|u^{(k)} - u^{(l)}\|_{\mathcal{H}}$, and has a limit $u_j \in H_j$. Choose a common bound M for the norms of the sequence. For every fixed N , passage to the coordinate limits gives $\sum_{j=1}^N \|u_j\|^2 \leq M^2$, so $u \in \mathcal{H}$. Given $\varepsilon > 0$, take K so that $\|u^{(k)} - u^{(l)}\|_{\mathcal{H}} \leq \varepsilon$ for $k, l \geq K$. Holding $k \geq K$ and N fixed, let $l \rightarrow \infty$, then take the supremum over N :

$$\sum_{j=1}^N \|u_j^{(k)} - u_j\|^2 \leq \varepsilon^2, \quad \|u^{(k)} - u\|_{\mathcal{H}} \leq \varepsilon. \quad (\text{HF21})$$

This proves completeness. Finite truncations converge to any u because the squared tail of its defining convergent series tends to zero. Deleting a set of coordinates is a bounded orthogonal projection with norm at most one, by the exact norm sum and pairing. Thus the direct sums used in the operator summation proof are actual Hilbert spaces, with their stated coordinate projections.

For a bounded $A : H \rightarrow K$, (HF4) gives its bounded adjoint and $\|A^*\| = \|A\|$. The original product $A^*A : H \rightarrow H$ obeys

$$\|A^*A\| \leq \|A^*\| \|A\| = \|A\|^2, \quad \|Ax\|^2 = (A^*Ax, x)_H \leq \|A^*A\| \|x\|^2. \quad (\text{HF22})$$

The pairing equality follows by conjugating $(Ax, Ax)_K = (x, A^*Ax)_H$; both diagonal values are real. Taking the supremum in the second inequality over $\|x\| \leq 1$ proves the reverse norm inequality. Consequently $\|A^*A\| = \|A\|^2$, including zero operators and zero spaces. If $A = A^*$ on one Hilbert space, this gives $\|A^2\| = \|A\|^2$. Applying it successively to the bounded self-adjoint operators $A, A^2, A^4, \dots$ gives $\|A^{2^k}\| = \|A\|^{2^k}$. No spectral theorem is needed for these norm identities.

Finally let $T : H \rightarrow H$ be bounded with $q = \|T\| < 1$. The space $\mathcal{L}(H)$ is complete in its operator norm: an operator-norm Cauchy sequence has a pointwise limit by completeness of H ; passage to the limit preserves linearity and the common norm bound. Passing its uniform Cauchy estimate to that pointwise limit and taking the unit-ball supremum proves operator-norm convergence. Composition is continuous because $\|AB\| \leq \|A\|\|B\|$.

The finite sums $S_N = \sum_{j=0}^N T^j$ are therefore Cauchy, since every finite tail is bounded by the corresponding scalar geometric tail. Their limit S satisfies

$$S = \sum_{j=0}^{\infty} T^j, \quad \|S - S_N\| \leq \frac{q^{N+1}}{1-q}, \quad \|S\| \leq \frac{1}{1-q}, \quad (I - T)S = S(I - T) = I. \quad (\text{HF23})$$

Both product equalities follow from the exact finite identities $(I - T)S_N = S_N(I - T) = I - T^{N+1}$, passage to the operator-norm limit, and $\|T^{N+1}\| \leq q^{N+1} \rightarrow 0$. The two products give both injectivity and surjectivity of $I - T$, so S is its unique inverse on all of H . The same proof works on any Banach space using its completeness. It proves the geometric inverse actually used in the divergence argument, without imposing a finite-dimensional restriction on that space or on the Hahn–Banach extension.

References and relation to the boundary course

The measure representation is the classical Riesz representation theorem on the compact interval. The functional-calculus route has the human antecedent Markus Haase, The Functional Calculus Approach to the Spectral Theorem, arXiv:2003.06130v2, especially the original TeX theorem labels `con.t.ext-bdd`, `con.t.ext-cont`, `spt.t.bdd`, and `spt.t.unb`.

For the differential-operator application, Dirichlet realizations, spectral projectors, and local extensions, Section 10 of Dirichlet realizations, spectral projectors, and local extensions, supplies the exact expression, weighted Hilbert space and boundary-domain inclusion. Section 4 proves the spectral facts on that unchanged Hilbert space. Its moment identity (LB14), recursive-power identity (LB16), closed endpoint (LB17), and explicit constant (LB19) provide precisely the spectral step in (DSP1) and (DSP40). The local differential regularity and kernel construction keep their complete proofs in that unit. No compactness or eigenbasis premise is added to that application.